

BANK LOAN DATA ANALYTICS SYSTEM

A DISSERTATION

**SUBMITTED TO
DEPARTMENT OF COMPUTER ENGINEERING AND
TECHNOLOGY**

GURU NANAK DEV UNIVERSITY, AMRITSAR

**FOR THE PARTIAL FULFILMENT OF THE REQUIREMENTS
FOR THE AWARD OF DEGREE**

OF

**BACHELOR OF TECHNOLOGY
(COMPUTER SCIENCE AND ENGINEERING)**

2022-2026

Supervised By:

Mr. Chetan Marwaha
Department of CET

Submitted By:

Manpreet Kaur
B.Tech (CSE)
Roll No.17032206339



**DEPARTMENT OF COMPUTER ENGINEERING AND
TECHNOLOGY**

GURU NANAK DEV UNIVERSITY

AMRITSAR

DECLARATION

I hereby declare that the Industrial Training Report entitled ("Bank Loan Data Analytics System ") is an authentic record of my own work as requirements of 6-months Industrial Training during the period from January, 2026 to May, 2026 for the award of degree of B. Tech. (CSE) from Department of Computer Engineering and Technology, Guru Nanak Dev University, Amritsar under the guidance of Mr Chetan Marwaha .

Date: 22 May 2026

Manpreet Kaur

Place: Amritsar

B.Tech (CSE)

Roll No.: 17032206339

This is to certify that above statement made by the candidate is correct and true to the best of my knowledge.

Mr Chetan Marwaha

Project Guide

(Signature and Seal)

Department of CET

ACKNOWLEDGEMENT

It is my pleasure to be indebted to various people who directly or indirectly contributed to the development of this project and influenced my thinking, behaviour, and actions during the course of study.

I express my sincere gratitude to all those who provided me with the opportunity to undertake this Major Project as a part of the academic curriculum.

I express my heartfelt gratitude to my project supervisor, **Mr. Chetan Marwaha**, Department of Computer Science and Engineering, for his invaluable guidance, constant encouragement, and unwavering support throughout the completion of this project.

I extend my sincere thanks to **Mr. Sandeep Sharma**, Head of the Department of Computer Science and Engineering, for providing the necessary academic resources, motivation, and a conducive environment for learning and research.

I would also like to acknowledge the contribution of various technologies and tools used during the project development, including PostgreSQL, Python, Jupyter Notebook, Microsoft Power BI, and Microsoft Excel, which greatly assisted in data analysis, visualization, and dashboard development.

Above all, I would like to sincerely thank God Almighty for His blessings, guidance, and support throughout this project. With His grace, I was able to complete this work successfully. I am truly grateful for the strength and opportunities He provided during this journey.

Finally, I express my deepest gratitude to my family and friends for their continuous motivation, encouragement, and emotional support throughout my academic journey and during the completion of this project.

Date: 22 May 2026

Place: Amritsar

Manpreet kaur

B.Tech CSE

Roll No.: 17032206339

ABSTRACT

The banking and financial sector generates vast volumes of loan data daily. Effectively harnessing this data is critical for operational monitoring, credit risk management, and strategic planning. This project presents the design and end-to-end implementation of a Bank Loan Data Analytics System — a multi-layered analytical solution that processes, analyses, and visualizes key loan performance indicators (KPIs) using industry-standard tools.

The project integrates four technology layers: (1) PostgreSQL for structured relational data storage and SQL-based KPI computation; (2) Python within Jupyter Notebook — using pandas, matplotlib, seaborn, and plotly.express — for comprehensive exploratory data analysis (EDA) producing 8 statistical charts; (3) Microsoft Power BI Desktop for 3-page interactive dashboard development with DAX-based time-intelligence measures; and (4) Microsoft Excel with pivot tables and Excel dashboards for summary and overview analysis.

The dataset comprises 38,576 loan records with 25 attributes. Key KPI results: Total Loan Applications (38,576 / MTD 4,314 / MoM +6.9%), Total Funded Amount (\$435.76M / MTD \$53.98M / MoM +13.0%), Total Amount Received (\$473.07M / MTD \$58.07M / MoM +15.8%), Average Interest Rate (12.05%), Average DTI (13.33%). Loan quality: Good Loans 86.18% (33,243 applications, \$370.22M funded) and Bad Loans 13.82% (5,333 applications, recovering only 56.9% of \$65.53M funded).

Three Power BI dashboards deliver Summary KPIs, Overview multi-dimensional analysis, and record-level Details. Excel dashboards provide Summary and Overview views with pivot-driven charts. Eight Python visualizations with full code are embedded in this report along with all dashboard screenshots.



SURE Trust - SKILL UPGRADATION FOR RURAL YOUTH EMPOWERMENT
Sreeguru Towers, Second Floor Gopuram Road, Opp. Union Bank of India
Puttaparthi. 515134 | +919440045717 / +917981538206 |
www.suretrustforruralityouth.com | Email: contact@suretrustforruralityouth.com



SURE Trust – AICTE Internship Offer Letter

Ms. Manpreet Kaur

B.Tech4th year
Guru Nanak Dev University, Amritsar
Amritsar, Punjab

Dear Manpreet Kaur,

SURE Trust is pleased to inform you that you are admitted as an Intern in the domain of “Data Science & Data Analytics” to work on industry-relevant projects for 6 months, starting from 01-02-2026. The management of SURE Trust has taken this decision after scrutinizing your technical skills in this domain. While there is no legal binding on either side in this contract, your commitment is taken as assured based on the words given by you. This internship requires you to work in complete coordination with the team, complete the assigned task by meeting the deadlines, and attend the weekly meetings mandatorily. You shall be reporting to Mr. Bhargavesh Dakka your mentor for your internship. It will be your responsibility to support the team in submitting the final report on project work completed in the required format and making it deployable. After completion of your internship, SURE Trust will give you an internship completion certificate after evaluating your performance. If this internship is acceptable to you kindly sign the letter and send it back for our records.

To
Ms. Manpreet Kaur

Prof. Radhakumari Challa
Founder & Executive Director, SURE Trust

Table of Contents

S.no.	Topic	Pages
1.	CHAPTER 1 INTRODUCTION	1
	1.1 Background and Motivation	1
	1.2 Problem Statement	1
	1.3 Project Objectives	1
2.	CHAPTER 2 LITERATURE REVIEW	2
	2.1 Business Intelligence in Financial Services	2
	2.2 Loan Default Prediction and Risk Analytics	2
	2.3 Data Visualization and Dashboard Design	3
	2.4 Gap Analysis and Contribution	3
3.	CHAPTER 3 SYSTEM REQUIREMENTS AND TECHNOLOGY STACK	4
	3.1 Functional Requirements	4
	3.2 Non-Functional Requirements	4
	3.3 Technology Stack	5
4.	CHAPTER 4 EXCEL DATA SOURCE AND DATA CLEANING	6
	4.1 Introduction	6
	4.2 Understanding the Raw Dataset	6
	4.3 Data Cleaning Process	7
	4.4 Clean Dataset	8
	4.5 Data Cleaning Summary	8
5.	CHAPTER 5 DATABASE DESIGN AND SQL IMPLEMENTATION	10
	5.1 Introduction	10
	5.2 Database Schema (25 Fields)	10
	5.3 SQL: Table Creation and Data Setup	11
	5.4 SQL: Total Loan Applications (Overall, MTD, PMTD)	13
	5.5 SQL: Total Funded Amount	14

	5.6 SQL: Total Amount Received	15
	5.7 SQL: Average Interest Rate	16
	5.8 SQL: Average DTI	17
	5.9 SQL: Good Loan KPIs	18
	5.10 SQL: Bad Loan KPIs	19
	5.11 SQL: Loan Status Grid View	20
	5.12 SQL: Overview Queries	21
	5.13 SQL Query Results Summary	26
6.	CHAPTER 6 PYTHON EXPLORATORY DATA ANALYSIS	27
	6.1 Introduction	27
	6.2 Libraries and Environment Setup	27
	6.3 Data Loading and Metadata Inspection	27
	6.4 KPI Calculations in Python	28
	6.5 Chart 1: Monthly Trends: Total Funded Amount	32
	6.6 Chart 2: Monthly Trends: Total Amount Received	33
	6.7 Chart 3: Monthly Trends: Total Loan Applications	34
	6.8 Chart 4: Funded Amount by State	36
	6.9 Chart 5: Loan Term Donut Chart	37
	6.10 Chart 6: Employee Length Bar Chart	38
	6.11 Chart 7: Loan Purpose Bar Chart	39
	6.12 Chart 8: Home Ownership Treemap	40
	6.13 Conclusion	40
7.	CHAPTER 7 MICROSOFT EXCEL DASHBOARD	41
	7.1 Introduction	41
	7.2 Excel Workbook Overview	41
	7.3 Design Sheet	42
	7.4 Summary Dashboard	42
	7.5 Overview Dashboard	45

	7.6 Detail page— Complete Loan Records	48
	7.7 Dashboard Features	49
	7.8 Conclusion	49
8.	CHAPTER 8 POWER BI DASHBOARD DESIGN AND IMPLEMENTATION	51
	8.1 Introduction to Power BI Dashboard	51
	8.2 Data Import and Preparation in Power BI	51
	8.3 DAX Measures Used in Power BI	52
	8.4 Power BI Dashboard Structure	53
	8.5 Summary Dashboard	53
	8.6 Overview Dashboard	54
	8.7 Details Dashboard	55
	8.8 Slicers and Interactivity	56
	8.9 Benefits of Power BI Dashboard	57
	8.10 Conclusion	57
9.	CHAPTER 9 RESULTS AND ANALYSIS	58
	9.1 Introduction	58
	9.2 Overall Loan Portfolio Analysis	58
	9.3 Good Loan vs Bad Loan Analysis	58
	9.4 Temporal and Geographic Insights	59
	9.5 Key Findings	59
	9.6 Conclusion	60
10.	CHAPTER 10 CONCLUSION AND FUTURE WORK	61
	10.1 Conclusion	61
	10.2 Limitations	61
	10.3 Future Work	62
11.	References	63

CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

The banking and financial sector is among the most data-intensive industries globally. Every loan application, disbursement, repayment, and default generates structured data that, when aggregated and analysed, reveals patterns critical to an institution's financial health. Traditional loan reporting relied on manual spreadsheet analysis and static periodic reports — an approach neither scalable nor responsive to modern business needs.

With cloud-hosted relational databases, scripting languages like Python, and self-service Business Intelligence platforms like Microsoft Power BI and Microsoft Excel, financial analysts now possess the capability to monitor loan portfolios dynamically, identify concentration risks, and measure performance against targets in real-time. This project demonstrates how a complete, end-to-end analytics pipeline can be built using industry-standard tools to serve a realistic banking analytics scenario.

1.2 Problem Statement

Banks generate enormous volumes of loan data but often lack consolidated, interactive systems to monitor portfolio performance at multiple levels of granularity. Specific operational problems motivating this system:

- Inability to track key KPIs (funded amounts, received amounts, interest rates, DTI) with MTD and MoM time breakdowns automatically.
- Difficulty distinguishing performing (Good) loans from non-performing (Bad/Charged-Off) loans systematically and at scale.
- Absence of multi-dimensional analysis across geography, term, employment profile, purpose, and home ownership.
- No consolidated interactive interface — from executive KPI summaries down to individual loan records.

1.3 Project Objectives

1. Design and implement a PostgreSQL relational database schema storing 38,576 loan records across 25 fields.

2. Develop 25+ SQL queries computing all KPIs with Overall, MTD, PMTD, and MoM percentage change.
3. Build Microsoft Excel dashboards (Summary and Overview) using pivot tables and Excel charts.
4. Perform full EDA using Python (pandas, matplotlib, seaborn, plotly) producing 8 labelled visualizations.
5. Build a 3-page interactive Power BI dashboard (Summary, Overview, Details) with DAX time-intelligence.
6. Cross-validate all KPIs across SQL, Python, Excel, and Power BI layers for accuracy.
7. Document all code, queries, charts, and dashboards completely in this project report.

CHAPTER 2: LITERATURE REVIEW

2.1 Business Intelligence in Financial Services

Business Intelligence (BI) helps banks and financial companies convert raw data into useful information for better decision-making. It uses tools and technologies to collect, organize, analyze, and present data in an easy-to-understand format.

In banking, BI is used to monitor customer information, track loan performance, identify risks, and improve operations. Dashboards and reports help managers quickly understand financial data and make better decisions

In this project, Business Intelligence tools such as SQL, Excel, Python, and Power BI are used to analyze bank loan data and create interactive dashboards for better understanding of loan performance

2.2 Loan Default Prediction and Risk Analytics

Loan default prediction helps banks identify customers who may fail to repay loans. Banks analyze factors such as income, employment history, loan amount, DTI ratio, and credit grade to measure risk.

Studies show that data analysis techniques improve loan risk assessment and help banks make better lending decisions. Proper data analysis and visualization are important before applying advanced prediction models.

This project focuses on descriptive analytics to study loan trends, repayment status, customer behavior, and risk distribution.

2.3 Data Visualization and Dashboard Design

Data visualization presents complex data in a simple and visual format using charts, graphs, and dashboards. Effective dashboards focus on important KPIs and help users easily understand business performance.

Different charts are used for different purposes:

- **Line Charts** → show trends over time
- **Bar Charts** → compare categories
- **Pie/Donut Charts** → show percentage distribution
- **Maps** → display geographic information

In this project, Excel and Power BI dashboards are used to display loan performance, funded amount, interest rates, and customer details in a clear and user-friendly way.

2.4 Gap Analysis and Contribution

Many existing studies focus only on loan prediction or dashboard visualization. This project combines complete data storage, SQL analysis, Python visualization, Excel reporting, and Power BI dashboards in one system.

The project includes:

- SQL database and queries
- KPI analysis
- Python data analysis
- Excel and Power BI dashboards
- Visual charts and reports

All codes, queries, charts, and dashboards are included in the report for easy understanding and future reference.

CHAPTER 3: SYSTEM REQUIREMENTS AND TECHNOLOGY STACK

3.1 Functional Requirements

The system should perform the following functions:

- Store more than 38,000 bank loan records safely in a PostgreSQL database.
- Calculate important KPIs such as total loan applications, funded amount, amount received, interest rate, and DTI ratio.
- Classify loans into:
 - **Good Loans** → Fully Paid and Current
 - **Bad Loans** → Charged Off
- Create dashboards in Excel and Power BI to display loan performance and customer insights.
- Analyze data based on different categories such as:
 - Monthly trends
 - State-wise analysis
 - Loan term
 - Employment length
 - Loan purpose
 - Home ownership
- Build an interactive Power BI dashboard with filters and slicers for easy analysis.

3.2 Non-Functional Requirements

The system should also meet the following quality requirements:

- **Performance:** SQL queries and dashboards should load quickly and work smoothly.
- **Accuracy:** KPI values should remain accurate across SQL, Python, Excel, and Power BI.
- **Usability:** Dashboards should be simple and easy to use for non-technical users.

- Scalability: The system should support a large amount of data for future expansion.

3.3 Technology Stack

Layer	Technology	Purpose
Data Storage	PostgreSQL	Store and manage bank loan data
DB Management	pgAdmin 4	Execute and manage SQL queries
Spreadsheet	Microsoft Excel	Create charts, pivot tables, and dashboards
Scripting	Python	Data cleaning, analysis, and visualization
IDE	Jupyter Notebook	Interactive Python coding environment
Data Manipulation	Pandas	Handle and manipulate data
Static Charts	Matplotlib	Line, bar, pie/donut, and area charts
Statistical Viz	Seaborn	Enhanced chart styling
Interactive Viz	plotly .express	Home ownership treemap
BI Dashboard	Power BI Desktop	Create interactive BI dashboards
Query Language	SQL (PostgreSQL)	Query and analyze database records
BI Formula Lang	DAX	Create KPIs and calculations in Power BI

CHAPTER 4: EXCEL DATA SOURCE AND DATA CLEANING

4.1 Introduction

The project started with a raw bank loan dataset available in an Excel file named **financial_loan.xlsx**. The dataset contained loan details of customers and was used as the main source for the entire project.

The Excel file contained **38,576 loan records** and **25 columns** with information such as:

- Loan Amount
- Interest Rate
- Loan Status
- Annual Income
- Employment Length
- Loan Purpose
- Home Ownership
- DTI Ratio

Initially, the data was unorganized and required cleaning before analysis and dashboard creation.

The **financial_loan** sheet is the single source of truth for all downstream analysis — SQL queries, Python EDA, Excel dashboards, and Power BI dashboards all derive their results from this dataset. Before any analysis could begin, the data was carefully inspected and cleaned to ensure accuracy, consistency, and completeness.

4.2 Understanding the Raw Dataset

The raw dataset was stored in the **financial_loan** sheet of the Excel file. This sheet contained all customer and loan information used later in SQL, Python, Excel, and Power BI.

While checking the raw data, some issues were identified:

- Missing values in the **emp_title** column
- Extra spaces in some columns

- Different date formats
- Need for proper loan classification

Because of these issues, data cleaning was necessary before performing analysis.

4.3 Data Cleaning Process

Data cleaning is the critical first step in any analytics project. Raw data often contains missing values, duplicates, formatting inconsistencies, and outliers that can distort analytical results. For this project, the following systematic data cleaning process was applied to the financial_loan dataset:

4.3.1 Step 1 — Check for Duplicate Records

The first step was to verify that no duplicate loan records exist. Using Excel Remove Duplicates, it was confirmed that zero duplicate rows exist. Each of the 38,576 records has a unique loan ID, confirming data integrity at the record level.

4.3.2 Step 2 — Identify and Handle Missing Values

The dataset was scanned for null/missing values across all 25 columns. Only one column — emp_title — had 1,438 missing records (3.73%) — the borrower job title field. This is expected in real-world loan data as not all borrowers disclose this. The null values were filled with “Not Specified” to maintain completeness without introducing bias.

4.3.3 Step 3 — Fix Data Type and Format Issues

Several formatting issues were identified and corrected:

- Extra spaces in the **term** column were removed using Trim() function
- Date columns were converted into proper date format
- Interest rate and DTI columns were formatted correctly for calculations and visualization

After cleaning, the data became consistent across all tools.

4.3.4 Step 4 — Check for Invalid/Outlier Values

Numeric columns were checked for logically invalid values: no negative loan amounts, no negative interest rates, no negative DTI values, and no annual incomes of zero were found. The minimum loan amount is \$500 and maximum is \$35,000, both within reasonable bounds for personal loans.

4.3.5 Step 5 — Add Derived Column

A new column "Good vs Bad Loan" was added to the Excel dataset using the following IF formula applied to all 38,576 rows:

```
=IF([@loan_status]="Charged Off", "Bad Loan", "Good Loan")
```

Logic:

"Charged Off" loans --> Bad Loan (5,333 records = 13.82%)

"Fully Paid" or "Current" loans --> Good Loan (33,243 records = 86.18%)

This derived column enables quick filtering in Excel slicers and Power BI slicers to separate performing from non-performing loans.

4.4 Clean Dataset

After completing all cleaning steps:

- Missing values were handled
- Formatting issues were corrected
- Duplicate records were removed
- Loan categories were properly classified

The cleaned dataset was now ready for analysis and dashboard creation.

4.5 Data Cleaning Summary

Check Performed	Findings (Before)	Action Taken	Result (After)
Duplicate records	0 duplicates found	No action needed	0 duplicates OK
Null values	1,438 nulls in emp_title	Filled with "Not Specified"	0 nulls in any column
Whitespace in term	Leading spaces in " 36 months"	Applied str.strip() / TRIM()	Clean "36 months" values

Data types	int_rate & dti as decimals	Multiply x100 in queries	Displayed as % correctly
Date formats	4 date columns mixed	Standardised to DATE type	Consistent across all tools
Negative values	Checked all numeric cols	None found	All numeric cols valid
loan_status values	3 unique values present	Confirmed valid categories	Fully Paid/Current/Charged Off
ID uniqueness	38,576 records	Verified no duplicates	38,576 unique IDs confirmed
Good vs Bad Loan	Column not present	Added IF formula in Excel	33,243 Good / 5,333 Bad

This chapter explained the complete process from receiving the raw Excel dataset to cleaning the data.

The cleaned and organized dataset became the foundation for:

- SQL analysis
- Python visualization
- Excel dashboards
- Power BI dashboards

Proper data cleaning improved the accuracy, consistency, and reliability of the final analytical results.

CHAPTER 5: DATABASE DESIGN AND SQL IMPLEMENTATION

5.1 Introduction

After cleaning the Excel dataset, the next step was to store the data in a database for analysis. PostgreSQL was used as the database management system because it helps store large amounts of data efficiently and allows fast data analysis using SQL queries.

The cleaned dataset containing **38,576 loan records** was imported into a table named **financial_loan**.

5.2 Database Schema (25 Fields)

A single table named **financial_loan** was created in PostgreSQL to store all loan-related information. The Excel file adds a calculated column "Good vs Bad Loan" making it 25 columns. This database design helps in easy data storage, fast querying, and dashboard creation.

#	Field	Data Type	Description
1	id	BIGINT (PK)	Unique loan identifier — primary key
2	address_state	VARCHAR(2)	US state code — regional risk assessment
3	application_type	VARCHAR(50)	Individual or Joint application
4	emp_length	VARCHAR(20)	Employment duration — repayment capacity
5	emp_title	VARCHAR(255)	Borrower job title — income verification
6	grade	VARCHAR(2)	Credit grade A–G — interest rate pricing driver
7	home_ownership	VARCHAR(20)	RENT / OWN / MORTGAGE status
8	issue_date	DATE	Loan origination date — MTD calculations
9	last_credit_pull_date	DATE	Latest credit inquiry date
10	last_payment_date	DATE	Most recent payment date
11	loan_status	VARCHAR(50)	Fully Paid / Current / Charged Off
12	Good vs Bad Loan	VARCHAR(20)	Derived: Good Loan / Bad Loan (Excel column)

13	next_payment_date	DATE	Next estimated payment date
14	member_id	BIGINT	Unique borrower identifier
15	purpose	VARCHAR(100)	Loan reason — product segmentation
16	sub_grade	VARCHAR(5)	Refined grade A1–G5 — fine rate pricing
17	term	VARCHAR(20)	36 or 60 months — repayment horizon
18	verification_status	VARCHAR(50)	Income verification outcome
19	annual_income	NUMERIC(15,2)	Annual income in USD
20	dti	NUMERIC(10,4)	Debt-to-income ratio (decimal)
21	installment	NUMERIC(10,2)	Fixed monthly payment (USD)
22	int_rate	NUMERIC(10,4)	Annual interest rate (decimal)
23	loan_amount	INT	Principal loan amount (USD)
24	total_acc	INT	Total credit accounts held
25	total_payment	NUMERIC(15,2)	Total amount received to date (USD)

5.3 SQL: Table Creation and Data Setup

SECTION 1: TABLE CREATION

```
DROP TABLE IF EXISTS financial_loan;
```

```
CREATE TABLE financial_loan (
```

```
id BIGINT PRIMARY KEY,
```

```
address_state VARCHAR(2),
```

```
application_type VARCHAR(50),
```

```
emp_length VARCHAR(20),
```

```
emp_title VARCHAR(255),
grade VARCHAR(2),
home_ownership VARCHAR(20),
issue_date DATE,
last_credit_pull_date DATE,
last_payment_date DATE,
loan_status VARCHAR(50),
next_payment_date DATE,
member_id BIGINT,
purpose VARCHAR(100),
sub_grade VARCHAR(5),
term VARCHAR(20),
verification_status VARCHAR(50),
annual_income NUMERIC(15, 2),
dti NUMERIC(10, 4),
installment NUMERIC(10, 2),
int_rate NUMERIC(10, 4),
loan_amount INT,
total_acc INT,
total_payment NUMERIC(15, 2)
);
SET datestyle = 'ISO, DMY';
SELECT * FROM financial_loan LIMIT 10;
```

5.4 SQL: Total Loan Applications (Overall, MTD, PMTD) Queries

Total Loan Applications (Overall)

```
SELECT COUNT(id) AS Total_Applications  
  
FROM financial_loan;
```

total_applications
38576

MTD Loan Applications(December)

```
SELECT COUNT(id) AS MTD_Total_Applications  
  
FROM financial_loan  
  
WHERE EXTRACT(MONTH FROM issue_date) = 12;
```

mtd_total_applications
4314

PMTD Loan Applications (November)

```
SELECT COUNT(id) AS PMTD_Total_Applications  
  
FROM financial_loan  
  
WHERE EXTRACT(MONTH FROM issue_date) = 11;
```

pmtd_total_applications
4035

5.5 SQL: Total Funded Amount (Overall, MTD, PMTD) Queries

Total Funded Amount

```
SELECT SUM(loan_amount) AS Total_Funded_Amount  
  
FROM financial_loan;
```

total_funded_amount
bigint
435757075

MTD Total Funded Amount

```
SELECT SUM(loan_amount) AS MTD_Total_Funded_Amount  
  
FROM financial_loan  
  
WHERE EXTRACT(MONTH FROM issue_date) = 12;
```

mtd_total_funded_amount
bigint
53981425

PMTD Total Funded Amount

```
SELECT SUM(loan_amount) AS PMTD_Total_Funded_Amount  
  
FROM financial_loan  
  
WHERE EXTRACT(MONTH FROM issue_date) = 11;
```

pmt_d_total_funded_amount
bigint
47754825

5.6 SQL: Total Amount Received (Overall, MTD, PMTD) Queries

Total Amount Received

```
SELECT SUM(total_payment) AS Total_Amount_Collected  
  
FROM financial_loan;
```

total_amount_collected
numeric
473070933.00

MTD Total Amount Received

```
SELECT SUM(total_payment) AS MTD_Total_Amount_Collected  
  
FROM financial_loan  
  
WHERE EXTRACT(MONTH FROM issue_date) = 12;
```

mtd_total_amount_collected
numeric
58074380.00

PMTD Total Amount Received

```
SELECT SUM(total_payment) AS PMTD_Total_Amount_Collected  
  
FROM financial_loan  
  
WHERE EXTRACT(MONTH FROM issue_date) = 11;
```

pmt_d_total_amount_collected
numeric
50132030.00

5.7 SQL: Average Interest Rate (Overall, MTD, PMTD) Queries

Average Interest Rate

```
SELECT AVG(int_rate)*100 AS Avg_Int_Rate  
  
FROM financial_loan;
```

avg_int_rate	
numeric	
12.04883139776026545000	

MTD Average Interest

```
SELECT AVG(int_rate)*100 AS MTD_Avg_Int_Rate  
  
FROM financial_loan  
  
WHERE EXTRACT(MONTH FROM issue_date) = 12;
```

mtd_avg_int_rate	
numeric	
12.35604079740380157600	

PMTD Average Interest

```
SELECT AVG(int_rate)*100 AS PMTD_Avg_Int_Rate  
  
FROM financial_loan  
  
WHERE EXTRACT(MONTH FROM issue_date) = 11;
```

pmtd_avg_int_rate	
numeric	
11.94171747211895910800	

5.8 SQL: Average DTI (Overall, MTD, PMTD) Queries

Avg DTI

```
SELECT AVG(dti)*100 AS Avg_DTI
```

```
FROM financial_loan;
```

avg_dti	
numeric	
13.32743311903774367500	

MTD Avg DTI

```
SELECT AVG(dti)*100 AS MTD_Avg_DTI
```

```
FROM financial_loan
```

```
WHERE EXTRACT(MONTH FROM issue_date) = 12;
```

mtd_avg_dti	
numeric	
13.66553778395920259600	

PMTD Avg DTI

```
SELECT AVG(dti)*100 AS PMTD_Avg_DTI
```

```
FROM financial_loan
```

```
WHERE EXTRACT(MONTH FROM issue_date) = 11;
```

pmtd_avg_dti	
numeric	
13.30273358116480793100	

5.9 SQL: Good Loan KPI Queries

Good Loans = loan_status IN ('Fully Paid', 'Current') — loans repaid or actively being repaid on schedule.

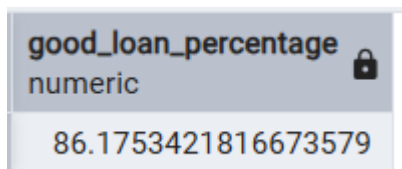
Good Loan Percentage

SELECT

(COUNT(CASE WHEN loan_status = 'Fully Paid' OR loan_status = 'Current' THEN id END) * 100.0) /

COUNT(id) AS Good_Loan_Percentage

FROM financial_loan;



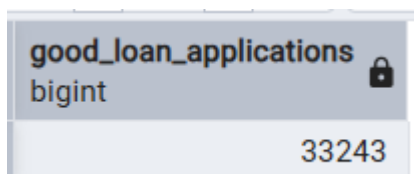
good_loan_percentage	numeric
86.1753421816673579	

Good Loan Applications

SELECT COUNT(id) AS Good_Loan_Applications

FROM financial_loan

WHERE loan_status = 'Fully Paid' OR loan_status = 'Current';



good_loan_applications	bigint
33243	

Good Loan Funded Amount

SELECT SUM(loan_amount) AS Good_Loan_Funded_amount

FROM financial_loan

WHERE loan_status = 'Fully Paid' OR loan_status = 'Current';

good_loan_funded_amount	bigint
	370224850

Good Loan Amount Received

```
SELECT SUM(total_payment) AS Good_Loan_amount_received
FROM financial_loan
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current';
```

good_loan_amount_received	numeric
	435786170.00

5.10 SQL: Bad Loan KPI Queries

Bad Loans = loan_status = 'Charged Off' — loans written off due to prolonged non-payment

Bad Loan Percentage

```
SELECT
    (COUNT(CASE WHEN loan_status = 'Charged Off' THEN id END) * 100.0) /
    COUNT(id) AS Bad_Loan_Percentage
FROM financial_loan
```

bad_loan_percentage	numeric
	13.8246578183326421

Bad Loan Applications

```
SELECT COUNT(id) AS Bad_Loan_Applications
FROM financial_loan
WHERE loan_status = 'Charged Off'
```

bad_loan_applications
bigint
5333

Bad Loan Funded Amount

```
SELECT SUM(loan_amount) AS Bad_Loan_Funded_amount
FROM financial_loan
WHERE loan_status = 'Charged Off'
```

bad_loan_funded_amount
bigint
65532225

Bad Loan Amount Received

```
SELECT SUM(total_payment) AS Bad_Loan_amount_received
FROM financial_loan
WHERE loan_status = 'Charged Off'
```

bad_loan_amount_received
numeric
37284763.00

5.11 SQL: Loan Status Grid View

```
SELECT
    loan_status,
    COUNT(id) AS Total_Loan_Applications,
    SUM(total_payment) AS Total_Amount_Received,
    SUM(loan_amount) AS Total_Funded_Amount,
    AVG(int_rate * 100) AS Interest_Rate,
```

AVG(dti * 100) AS DTI

FROM financial_loan

GROUP BY loan_status;

loan_status character varying (50)	total_loan_applications bigint	total_amount_received numeric	total_funded_amount bigint	interest_rate numeric	dti numeric
Current	1098	24199914.00	18866500	15.0993260473588342	14.7243442622950820
Fully Paid	32145	411586256.00	351358350	11.6410707730595738	13.1673507543941515
Charged Off	5333	37284763.00	65532225	13.8785749109319332	14.0047327957997375

MTD breakdown (December)

SELECT

loan_status,

SUM(total_payment) AS MTD_Total_Amount_Received,

SUM(loan_amount) AS MTD_Total_Funded_Amount

FROM financial_loan

WHERE EXTRACT(MONTH FROM issue_date) = 12

GROUP BY loan_status;

loan_status character varying (50)	mtd_total_amount_received numeric	mtd_total_funded_amount bigint
Charged Off	5324211.00	8732775
Current	4934318.00	3946625
Fully Paid	47815851.00	41302025

5.12 SQL: Overview Queries (6 Analytical Dimensions)

5.12.1 Monthly Trend

SELECT

EXTRACT(MONTH FROM issue_date) AS Month_Number,

TO_CHAR(issue_date, 'Month') AS Month_name,

```

COUNT(id) AS Total_Loan_Applications,

SUM(loan_amount) AS Total_Funded_Amount,

SUM(total_payment) AS Total_Amount_Received

FROM financial_loan

GROUP BY Month_Number, Month_name

ORDER BY Month_Number;

```

month_number numeric	month_name text	total_loan_applications bigint	total_funded_amount bigint	total_amount_received numeric
1	January	2332	25031650	27578836.00
2	February	2279	24647825	27717745.00
3	March	2627	28875700	32264400.00
4	April	2755	29800800	32495533.00
5	May	2911	31738350	33750523.00
6	June	3184	34161475	36164533.00
7	July	3366	35813900	38827220.00
8	August	3441	38149600	42682218.00
9	September	3536	40907725	43983948.00
10	October	3796	44893800	49399567.00
11	November	4035	47754825	50132030.00
12	December	4314	53981425	58074380.00

5.12.2 State Analysis

```

SELECT

address_state AS State,

COUNT(id) AS Total_Loan_Applications,

SUM(loan_amount) AS Total_Funded_Amount,

SUM(total_payment) AS Total_Amount_Received

FROM financial_loan

GROUP BY address_state

ORDER BY address_state;

```

state character varying (2)	total_loan_applications bigint	total_funded_amount bigint	total_amount_received numeric
AK	78	1031800	1108570.00
AL	432	4949225	5492272.00
AR	236	2529700	2777875.00
AZ	833	9206000	10041986.00
CA	6894	78484125	83901234.00
CO	770	8976000	9845810.00
CT	730	8435575	9357612.00
DC	214	2652350	2921854.00
DE	110	1138100	1269136.00
FL	2773	30046125	31601905.00
GA	1355	15480325	16728040.00
HI	170	1850525	2080184.00
IA	5	56450	64482.00
ID	6	59750	65329.00

5.12.3 Term Analysis

SELECT

term AS Term,

COUNT(id) AS Total_Loan_Applications,

SUM(loan_amount) AS Total_Funded_Amount,

SUM(total_payment) AS Total_Amount_Received

FROM financial_loan

GROUP BY term

ORDER BY term;

term character varying (20)	total_loan_applications bigint	total_funded_amount bigint	total_amount_received numeric
36 months	28237	273041225	294709458.00
60 months	10339	162715850	178361475.00

5.12.4 Employee Length Analysis

SELECT

emp_length AS Employee_Length,

COUNT(id) AS Total_Loan_Applications,

```

SUM(loan_amount) AS Total_Funded_Amount,

SUM(total_payment) AS Total_Amount_Received

FROM financial_loan

GROUP BY emp_length

ORDER BY emp_length;

```

employee_length character varying (20) 🔒	total_loan_applications bigint 🔒	total_funded_amount bigint 🔒	total_amount_received numeric 🔒
< 1 year	4575	44210625	47545011.00
1 year	3229	32883125	35498348.00
10+ years	8870	116115950	125871616.00
2 years	4382	44967975	49206961.00
3 years	4088	43937850	47551832.00
4 years	3428	37600375	40964850.00
5 years	3273	36973625	40397571.00
6 years	2228	25612650	27908658.00
7 years	1772	20811725	22584136.00
8 years	1476	17558950	19025777.00
9 years	1255	15084225	16516173.00

5.12.5 Home Ownership Analysis

```

SELECT

home_ownership AS Home_Ownership,

COUNT(id) AS Total_Loan_Applications,

SUM(loan_amount) AS Total_Funded_Amount,

SUM(total_payment) AS Total_Amount_Received

FROM financial_loan

GROUP BY home_ownership

ORDER BY home_ownership;

```


home_ownership character varying (20) 🔒	total_loan_applications bigint 🔒	total_funded_amount bigint 🔒	total_amount_received numeric 🔒
MORTGAGE	17198	219329150	238474438.00
NONE	3	16800	19053.00
OTHER	98	1044975	1025257.00
OWN	2838	29597675	31729129.00
RENT	18439	185768475	201823056.00

5.12.6 Loan Purpose Analysis

SELECT

purpose AS PURPOSE,

COUNT(id) AS Total_Loan_Applications,

SUM(loan_amount) AS Total_Funded_Amount,

SUM(total_payment) AS Total_Amount_Received

FROM financial_loan

GROUP BY purpose

ORDER BY purpose;

purpose character varying (100) 🔒	total_loan_applications bigint 🔒	total_funded_amount bigint 🔒	total_amount_received numeric 🔒
car	1497	10223575	11324914.00
credit card	4998	58885175	65214084.00
Debt consolidation	18214	232459675	253801871.00
educational	315	2161650	2248380.00
home improvement	2876	33350775	36380930.00
house	366	4824925	5185538.00
major purchase	2110	17251600	18676927.00
medical	667	5533225	5851372.00
moving	559	3748125	3999899.00
other	3824	31155750	33289676.00
renewable_energy	94	845750	898931.00
small business	1776	24123100	23814817.00
vacation	352	1967950	2116738.00
wedding	928	9225800	10266856.00

5.13 SQL Query Results Summary

KPI	Overall	MTD (Dec)	PMTD (Nov)	MoM Change
Total Loan Applications	38,576	4,314	4,035	+6.9%
Total Funded Amount	\$435.76M	\$53.98M	\$47.75M	+13.0%
Total Amount Received	\$473.07M	\$58.07M	\$50.13M	+15.8%
Average Interest Rate	12.05%	12.36%	11.94%	+3.5%
Average DTI	13.33%	13.67%	13.30%	+2.7%

Loan Category Analysis

Category	Applications	Funded Amount	Amount Received	Recovery Rate
Good (Fully Paid + Current)	33,243 (86.18%)	\$370.22M	\$435.79M	117.7%
Bad (Charged Off)	5,333 (13.82%)	\$65.53M	\$37.28M	56.9%

CHAPTER 6: PYTHON EXPLORATORY DATA ANALYSIS

6.1 Introduction

After completing SQL analysis, Python was used for Exploratory Data Analysis (EDA) and data visualization. Python helps analyze large datasets easily and create charts for better understanding of business insights.

The analysis was performed using Jupyter Notebook.

6.2 Libraries and Environment Setup

The following Python libraries were used in this project:

```
import pandas as pd          # Data manipulation
import numpy as np           # Numerical operations
import matplotlib.pyplot as plt # Charts and graphs
import seaborn as sns        # Statistical visualization
import warnings              # Ignore warnings
import plotly.express as px   # Interactive charts
```

Purpose of Libraries

- **Pandas** → Data cleaning and analysis
- **NumPy** → Mathematical calculations
- **Matplotlib & Seaborn** → Data visualization
- **Plotly** → Interactive charts and treemaps

6.3 Data Loading and Metadata Inspection

The Excel dataset was loaded into Python using Pandas.

```
df = pd.read_excel('financial_loan_data_excel.xlsx')
```

The dataset was checked using:

```
df.head()                                # Preview first 5 rows
df.tail()                                # Preview last 5 rows
print('NO. of Rows:', df.shape[0])
print('NO. of Columns:', df.shape[1])
df.dtypes                                # data types
df.describe()                            #statistical summary
```

OUTPUT:

NO. of Rows: 38576

NO. of Columns: 2

This confirmed that the cleaned dataset was ready for analysis.

6.4 KPI Calculations in Python

6.4.1 Total Loan Applications

```
total_loan_application = df['id'].count()
print("Total Loan Applications:", total_loan_application)
```

OUTPUT: Total Loan Applications: 38576

6.4.2 MTD Loan Applications

```
latest_issue_date = df['issue_date'].max()
latest_year = latest_issue_date.year
latest_month = latest_issue_date.month
mtd_data = df[(df['issue_date'].dt.year==latest_year) & (df['issue_date'].dt.month==latest_month)]
mtd_loan_applications = mtd_data['id'].count()
print(f'MTD Loan Applications (for {latest_issue_date.strftime('%B %Y')}): {mtd_loan_applications}')
")
```

OUTPUT: MTD Loan Applications (for December 2021): 4314

6.4.3 Total and MTD Funded Amount

```
total_funded_amount = df['loan_amount'].sum()

total_funded_amount_millions = total_funded_amount/ 1000000

print("Total Funded Amount:${:.2f}M".format(total_funded_amount_millions))
```

OUTPUT: Total Funded Amount: \$435.76M

```
latest_issue_date = df['issue_date'].max()

latest_year = latest_issue_date.year

latest_month = latest_issue_date.month

mtd_data = df[(df['issue_date'].dt.year == latest_year) & (df['issue_date'].dt.month == latest_month)]

mtd_total_funded_amount = mtd_data['loan_amount'].sum()

mtd_total_funded_amount_millions = mtd_total_funded_amount/ 1000000

print("MTD Total Funded Amount:${:.2f}M".format(mtd_total_funded_amount_millions))
```

OUTPUT: MTD Total Funded Amount: \$53.98M

6.4.4 Total and MTD Amount Received

```
total_amount_received = df['total_payment'].sum()

total_amount_received_millions = total_amount_received/ 1000000

print("Total Amount Recieved:${:.2f}M".format(total_amount_received_millions))
```

OUTPUT: Total Amount Received: \$473.07M

```
latest_issue_date = df['issue_date'].max()

latest_year = latest_issue_date.year

latest_month = latest_issue_date.month

mtd_data = df[(df['issue_date'].dt.year == latest_year) & (df['issue_date'].dt.month == latest_month)]

mtd_total_amount_received = mtd_data['total_payment'].sum()
```

```
mtd_total_amount_received_millions = mtd_total_amount_received/ 1000000

print("MTD Total Amount Recieved:${:.2f}M".format(mtd_total_amount_received_millions))
```

OUTPUT: MTD Total Amount Received: \$58.07M

6.4.5 Average Interest Rate and DTI

```
average_interest_rate = df['int_rate'].mean()*100

print("Avg Int Rate: {:.2f}%".format(average_interest_rate))
```

OUTPUT: Avg Int Rate: 12.05%

```
average_dti = df['dti'].mean()*100

print("Avg DTI: {:.2f}%".format(average_dti))
```

OUTPUT: Avg DTI: 13.33%

6.4.6 Good Loan Metrics

```
good_loans = df[df['loan_status'].isin(["Fully Paid", "Current"])]

total_loan_applications = df['id'].count()

good_loan_applications = good_loans['id'].count()

good_loan_funded_amount = good_loans['loan_amount'].sum()

good_loan_received = good_loans['total_payment'].sum()

good_loan_funded_amount_millions = good_loan_funded_amount/1000000

good_loan_received_millions = good_loan_received/1000000

good_loan_percentage=(good_loan_applications / total_loan_applications )*100

print ("Good Loan Applications:",good_loan_applications)

print ("Good Loan Funded Amount (in Millions): ${:.2f} M".format (good_loan_funded_ amount_
millions))

print("Good Loan Total Received ( in Millions):${:.2f}M".format(good_loan_received_millions))
```

```
print("Percentage of Good Loan Applications:${:.2f}%".format(good_loan_percentage))
```

OUTPUT: Good Loan Applications: 33243

Good Loan Funded Amount: \$370.22M

Good Loan Total Received: \$435.79M

Percentage of Good Loans: 86.18%

6.4.7 Bad Loan Metrics

```
bad_loans = df[df['loan_status'].isin(['Charged Off'])]
```

```
total_loan_applications = df['id'].count()
```

```
bad_loan_applications = bad_loans['id'].count()
```

```
bad_loan_funded_amount = bad_loans['loan_amount'].sum()
```

```
bad_loan_received = bad_loans['total_payment'].sum()
```

```
bad_loan_funded_amount_millions = bad_loan_funded_amount/1_000_000
```

```
bad_loan_received_millions = bad_loan_received/1_000_000
```

```
bad_loan_percentage=(bad_loan_applications / total_loan_applications )*100
```

```
print("Bad Loan Applications:",bad_loan_applications)
```

```
print("Bad Loan Funded Amount ( in Millions):${:.2f}M".format(bad_loan_funded_amount_millions))
```

```
print("Bad Loan Total Received ( in Millions):${:.2f}M".format(bad_loan_received_millions))
```

```
print("Percentage of Bad Loan Applications:${:.2f}%".format(bad_loan_percentage))
```

OUTPUT: Bad Loan Applications: 5333

Bad Loan Funded Amount: \$65.53M

Bad Loan Total Received: \$37.28M

Percentage of Bad Loans: 13.82%

6.5 Chart 1 — Monthly Trends: Total Funded Amount

```
monthly_funded = (df.sort_values('issue_date')

    .assign(month_name=lambda x: x['issue_date'].dt.strftime('%b %Y'))

    .groupby('month_name', sort=False)['loan_amount']

    .sum()

    .div(1000000)

    .reset_index(name='loan_amount_millions')

)

plt.figure(figsize=(10, 5))

plt.fill_between(monthly_funded['month_name'],monthly_funded['loan_amount_millions'], color=
'skyblue', alpha=0.5)

plt.plot(monthly_funded['month_name'], monthly_funded['loan_amount_millions'], color='blue',
linewidth =2)

for i, row in monthly_funded.iterrows():

    plt.text(i, row['loan_amount_millions'] + 0.1, f'{row["loan_amount_millions"]:.2f}',

            ha='center', va='bottom', fontsize=9, rotation=0, color='black')

plt.title('Total Funded Amount by Month', fontsize=14)

plt.xlabel('Month')

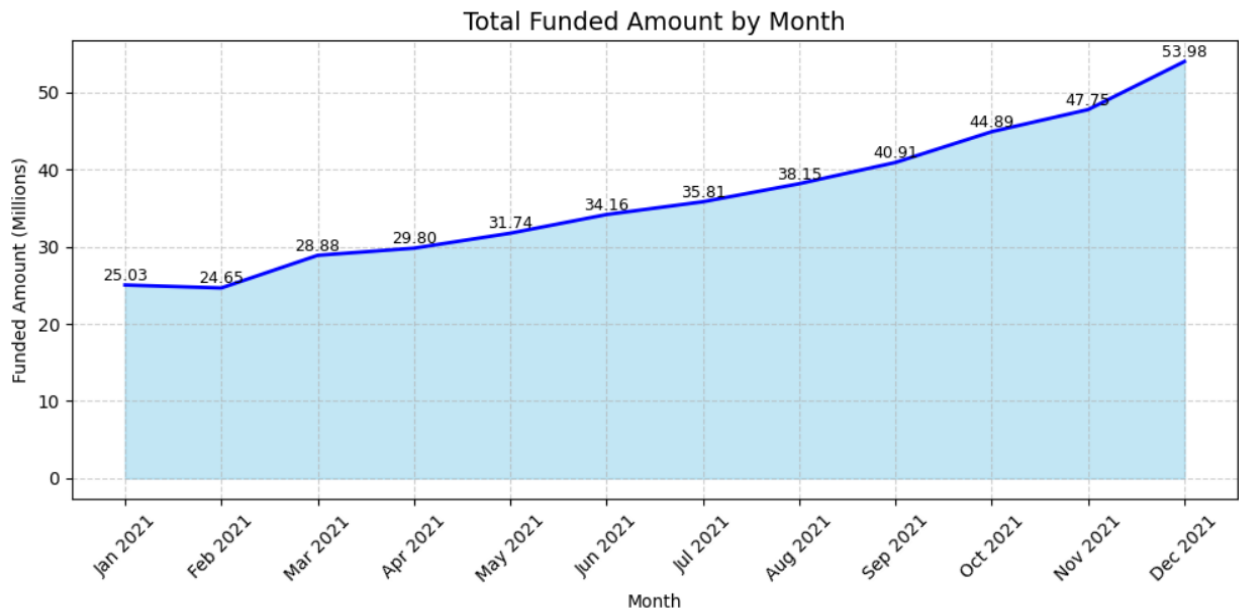
plt.ylabel('Funded Amount (Millions)')

plt.grid(True, linestyle='--', alpha=0.6)

plt.xticks(ticks=range(len(monthly_funded)), labels=monthly_funded['month_name'], rotation=45)

plt.tight_layout()

plt.show()
```

6.6 Chart 2 — Monthly Trends: Total Amount Received

```
monthly_received = (
    df.sort_values('issue_date')
        .assign(month_name=lambda x: x['issue_date'].dt.strftime('%b %Y'))
        .groupby('month_name', sort=False)['total_payment']
        .sum()
        .div(1_000_000)
        .reset_index(name='received_amount_millions')
)

plt.figure(figsize=(10, 5))

plt.fill_between(monthly_received['month_name'], monthly_received['received_amount_millions'],
                 color='lightgreen', alpha=0.5)

plt.plot(monthly_received['month_name'], monthly_received['received_amount_millions'],
         color='green', linewidth=2)
```

```

for i, row in monthly_received.iterrows():

    plt.text(i, row['received_amount_millions'] + 0.1, f'{row["received_amount_millions"]:.2f}',

             ha='center', va='bottom', fontsize=9, rotation=0, color='black')

plt.title("Total Received Amount by Month", fontsize=14)

plt.xlabel('Month')

plt.ylabel('Received Amount (Millions)')

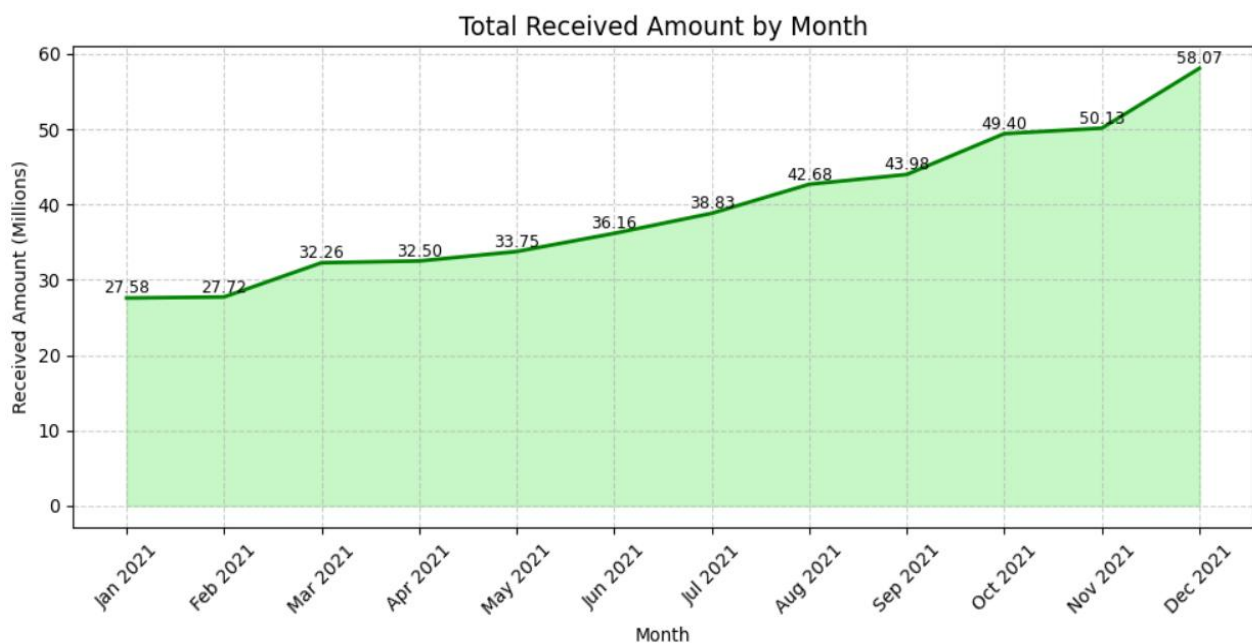
plt.xticks(ticks=range(len(monthly_received)), labels=monthly_received['month_name'], rotation=45)

plt.grid(True, linestyle='--', alpha=0.6)

plt.tight_layout()

plt.show()

```



6.7 Chart 3 — Monthly Trends: Total Loan Applications

```

monthly_applications = (

    df.sort_values('issue_date')

    .assign(month_name=lambda x: x['issue_date'].dt.strftime('%b %Y'))

    .groupby('month_name', sort=False)['id']

```

```

.count()

.reset_index(name='loan_applications_count')

)

plt.figure(figsize=(10, 5))

plt.fill_between(monthly_applications['month_name'],monthly_applications

                ['loan_applications_count'], color='orange', alpha=0.5)

plt.plot(monthly_applications['month_name'], monthly_applications['loan_applications_count'],

        color='darkorange', linewidth=2)

for i, row in monthly_applications.iterrows():

    plt.text(i, row['loan_applications_count'] + 0.5, f'{row["loan_applications_count"]}',

            ha='center', va='bottom', fontsize=9, rotation=0, color='black')

plt.title("Total Loan Applications by Month", fontsize=14)

plt.xlabel('Month')

plt.ylabel('Number of Applications')

plt.xticks(ticks=range(len(monthly_applications)),          labels=monthly_applications['month_name'],

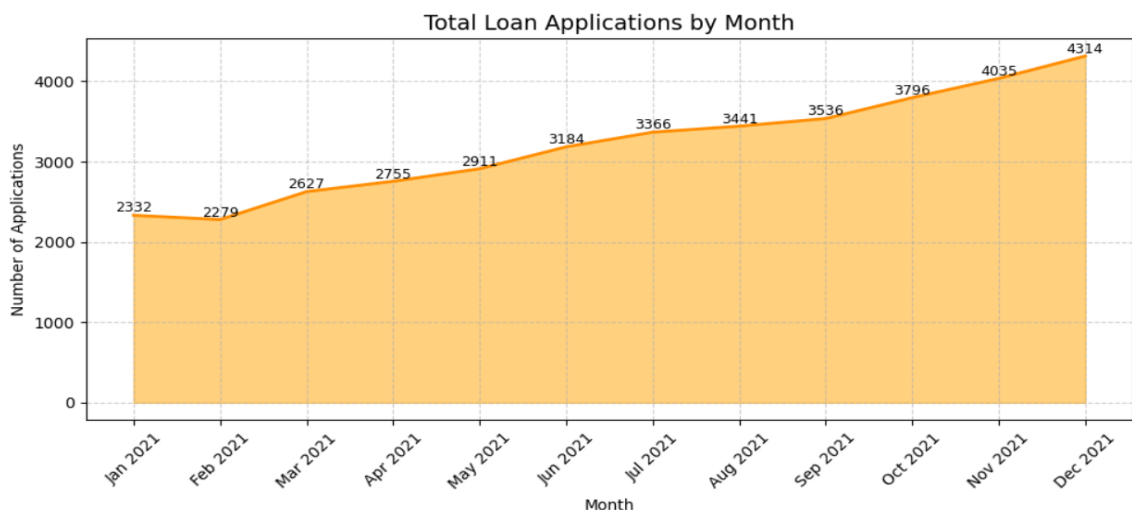
rotation=45)

plt.grid(True, linestyle='--', alpha=0.6)

plt.tight_layout()

plt.show()

```



6.8 Chart 4 — Regional Analysis: Funded Amount by State

```
state_funding = df.groupby('address_state')['loan_amount'].sum().sort_values(ascending=True)

state_funding_thousands = state_funding / 1000

plt.figure(figsize=(10, 8))

bars = plt.barh(state_funding_thousands.index, state_funding_thousands.values, color='lightcoral')

for bar in bars:

    width = bar.get_width()

    plt.text(width + 10, bar.get_y() + bar.get_height() / 2,

             f'{width:,.0f}K', va='center', fontsize=9)

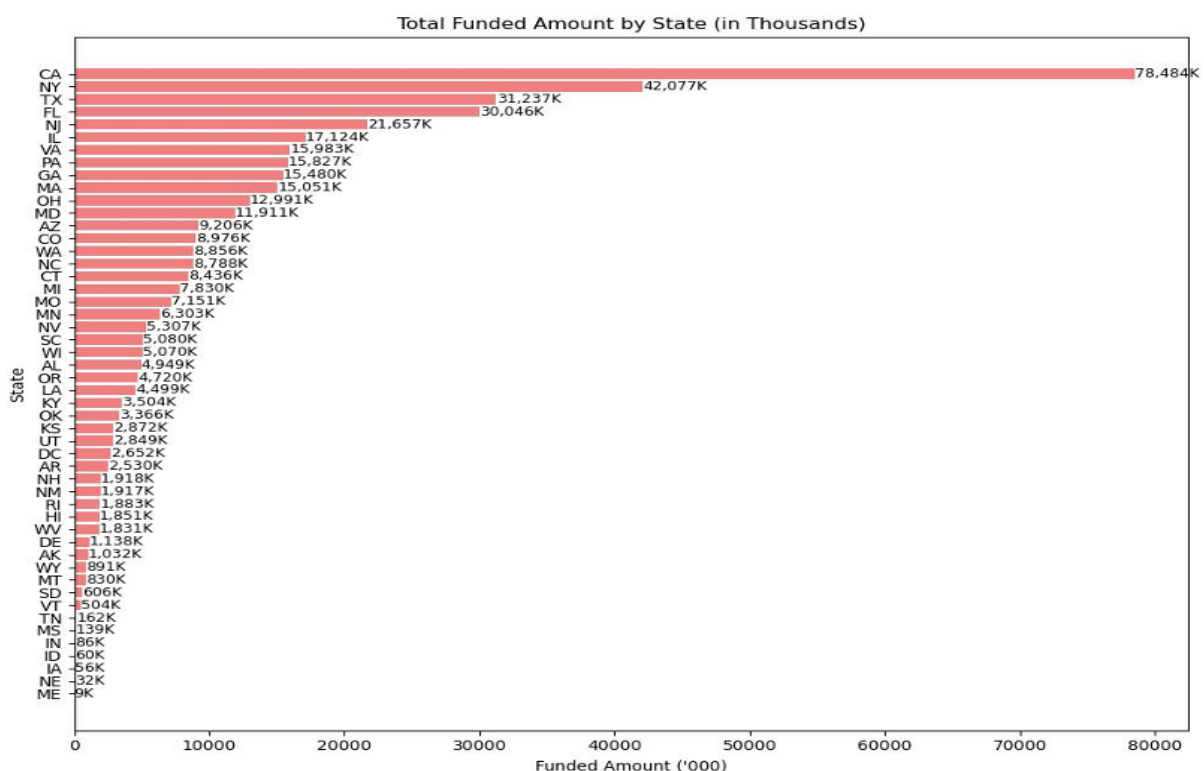
plt.title('Total Funded Amount by State (in Thousands)')

plt.xlabel('Funded Amount (\'000)')

plt.ylabel('State')

plt.tight_layout()

plt.show()
```



6.9 Chart 5 — Loan Term Analysis: Donut Chart

```
term_funding_millions = df.groupby('term')['loan_amount'].sum() / 1000000

plt.figure(figsize=(5,5))

plt.pie(

    term_funding_millions,

    labels=term_funding_millions.index,

    autopct=lambda p: f'{p:.1f}%\n${p*sum(term_funding_millions)/100:.1f}M',

    startangle=90,

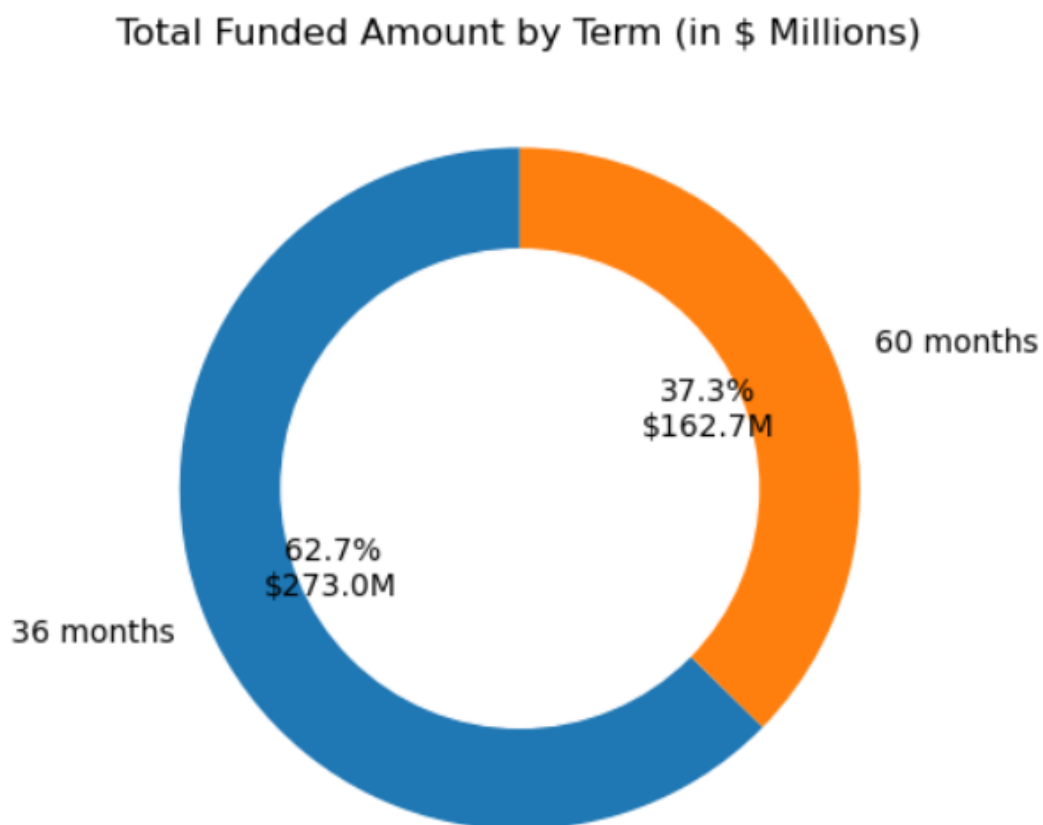
    wedgeprops={'width': 0.4}

)

plt.gca().add_artist(plt.Circle((0, 0), 0.70, color='white'))

plt.title("Total Funded Amount by Term (in $ Millions)")

plt.show()
```



6.10 Chart 6 — Employee Length: Total Funded Amount

```
emp_funding = df.groupby('emp_length')['loan_amount'].sum().sort_values()/1000
```

```
plt.figure(figsize=(10, 6))
```

```
bars = plt.barh(emp_funding.index, emp_funding, color='purple')
```

for bar in bars:

```
    width = bar.get_width()
```

```
    plt.text(width + 5, bar.get_y() + bar.get_height() / 2,
```

```
            f'{width:,.0f}K', va='center', fontsize=9)
```

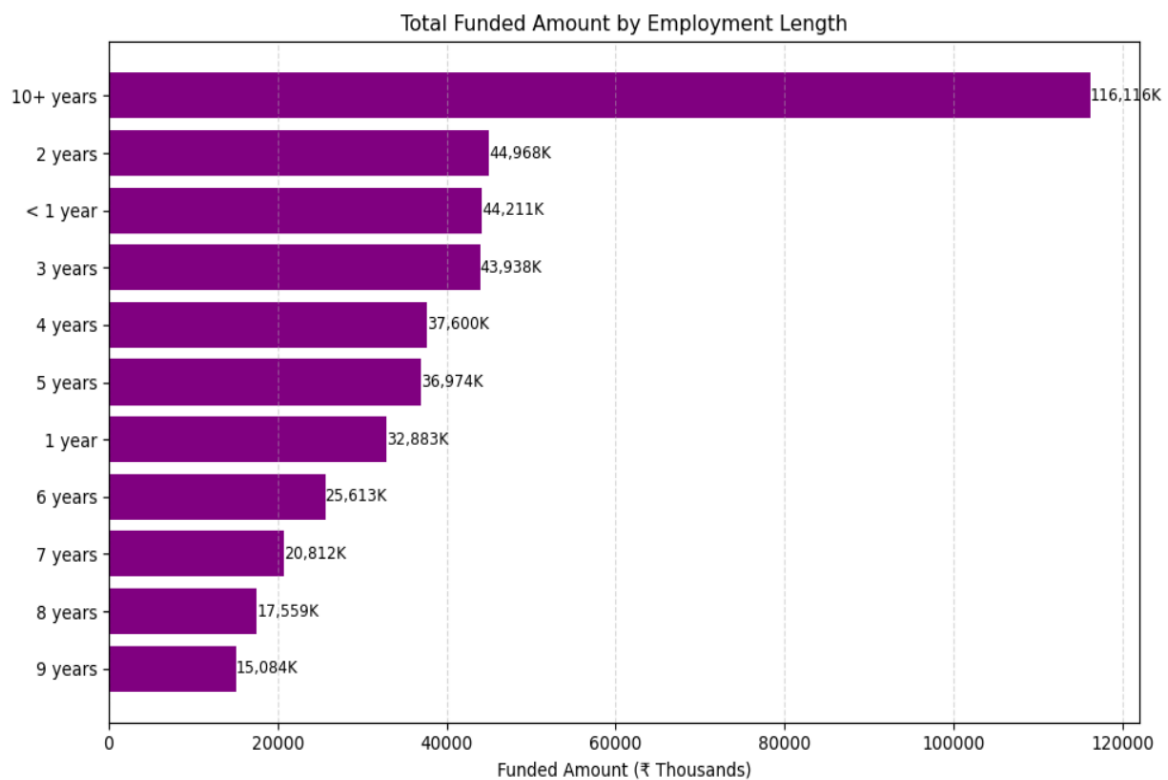
```
plt.xlabel("Funded Amount (₹ Thousands)")
```

```
plt.title("Total Funded Amount by Employment Length")
```

```
plt.grid(axis='x', linestyle='--', alpha=0.5)
```

```
plt.tight_layout()
```

```
plt.show()
```



6.11 Chart 7 — Loan Purpose: Total Funded Amount

```
purpose_funding_millions = (df.groupby('purpose')['loan_amount'].sum().sort_values() / 1000000)

plt.figure(figsize=(10, 6))

bars = plt.barh(purpose_funding_millions.index, purpose_funding_millions.values, color='skyblue')

for bar in bars:

    width = bar.get_width()

    plt.text(width + 0.1, bar.get_y() + bar.get_height() / 2,

             f'{width:.2f}M', va='center', fontsize=9)

plt.title('Total Funded Amount by Loan Purpose (Millions)', fontsize=14)

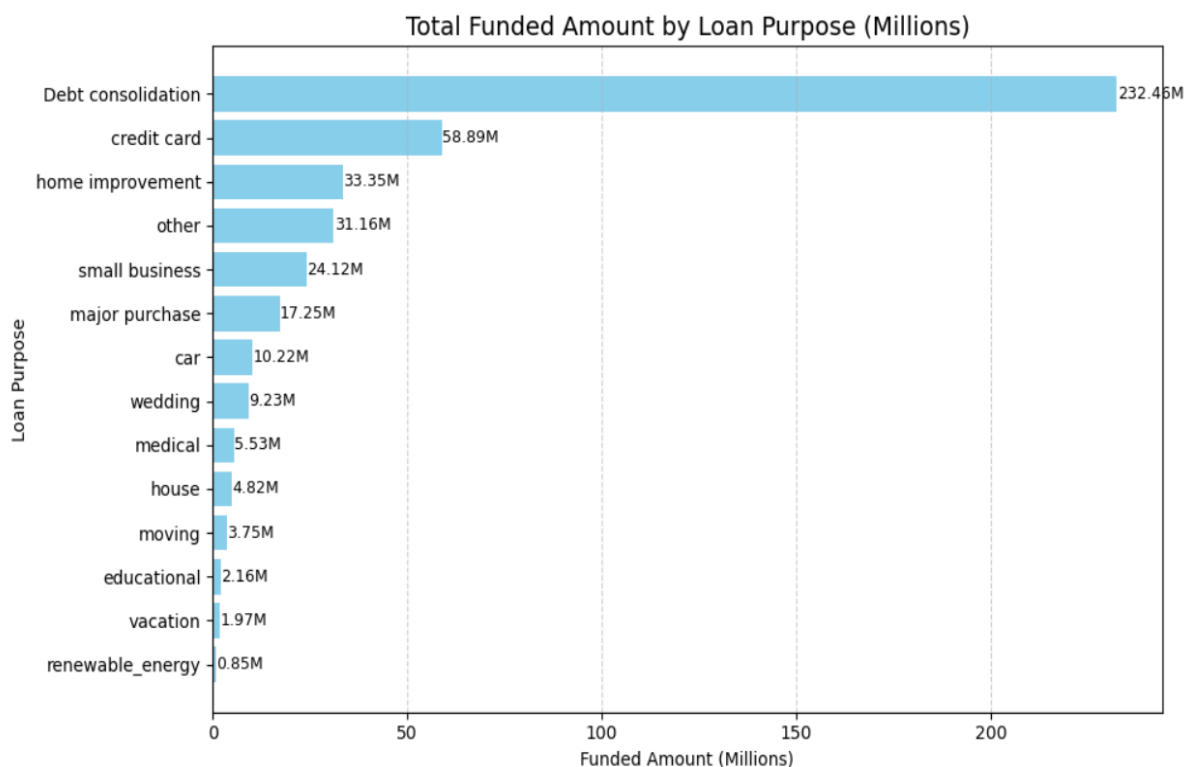
plt.xlabel('Funded Amount (Millions)')

plt.ylabel('Loan Purpose')

plt.grid(axis='x', linestyle='--', alpha=0.6)

plt.tight_layout()

plt.show()
```



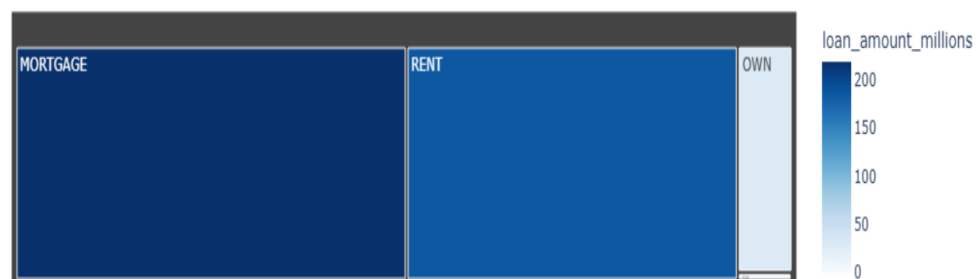
6.12 Chart 8 — Home Ownership: Treemap (Plotly Express)

```
home_funding = df.groupby('home_ownership')['loan_amount'].sum().reset_index()

home_funding['loan_amount_millions'] = home_funding['loan_amount'] / 1000000

fig = px.treemap(
    home_funding,
    path=['home_ownership'],
    values='loan_amount_millions',
    color='loan_amount_millions',
    color_continuous_scale='Blues',
    title='Total Funded Amount by Home Ownership (₹ Millions)'
)
fig.show()
```

Total Funded Amount by Home Ownership (₹ Millions)



6.13 Conclusion

Python Exploratory Data Analysis helped identify important loan trends, customer behavior, and repayment insights using charts and visualizations. These findings were later used in Excel and Power BI dashboards for interactive reporting and business analysis. The analysis provided:

- Better understanding of loan performance
- Identification of good and bad loans
- Monthly and regional trends
- Customer segmentation insights

CHAPTER 7: MICROSOFT EXCEL DASHBOARD

7.1 Introduction

After completing data cleaning, SQL analysis, and Python Exploratory Data Analysis (EDA), interactive dashboards were created in Microsoft Excel to visualize the bank loan dataset.

The Excel dashboard helps users understand loan performance, customer behavior, and financial trends in a simple and interactive way. The dashboards were designed using:

- Pivot Tables
- Pivot Charts
- Slicers
- Conditional Formatting
- KPI Cards

The dashboards provide a complete Business Intelligence solution inside Excel without using any external tool.

7.2 Excel Workbook Overview

The Microsoft Excel file `financial_loan_data_excel.xlsx` contains four distinct sheets structured to handle raw data, backend processing, and executive reporting:

- **financial_loan**: The raw data repository containing **38,576 rows** and **25 columns** of loan records.
- **Design Sheet**: The central calculation engine containing Pivot Table outputs and Key Performance Indicator (KPI) backing data.
- **SUMMARY DASHBOARD**: An executive-level reporting interface focusing on portfolio health and loan quality.
- **OVERVIEW DASHBOARD**: A multi-dimensional analysis interface focusing on borrower demographics, geographic distribution, and loan characteristics.

7.3 Design Sheet — Pivot Table Calculations

The Design Sheet acts as the backend calculation area for the dashboards. It contains multiple Pivot Tables used for KPI calculations and charts.

The pivot tables were used for:

- Loan Status Analysis
- Funded Amount Analysis
- Amount Received Analysis
- Interest Rate and DTI Analysis
- Monthly Trend Analysis
- Employee Length Analysis
- Loan Purpose Analysis
- Home Ownership Analysis

All dashboard charts and KPI cards are dynamically linked to these pivot tables.

When slicers are changed, all charts and values automatically update.

7.4 Summary Dashboard

The **SUMMARY DASHBOARD** provides a high-level overview of the entire loan portfolio.

The dashboard is divided into three sections:

1. KPI Cards
2. Good Loan vs Bad Loan Analysis
3. Loan Status Breakdown

7.4.1 KPI Cards

The top section displays important KPI metrics.

KPI Metric	Value
Total Loan Applications	38.6K
Total Funded Amount	\$435.8M
Total Amount Received	\$473.1M
Average Interest Rate	12.05%
Average DTI	13.33%

The dashboard also shows:

- MTD (Month-To-Date) values
- MoM (Month-over-Month) growth percentage

These KPI cards help users quickly monitor loan performance.

7.4.2 Good Loan vs Bad Loan Analysis

The middle section contains donut charts showing loan quality distribution.

Good Loans

- Applications: 33,243
- Percentage: 86.18%
- Funded Amount: \$370.2M
- Amount Received: \$435.8M

Bad Loans

- Applications: 5,333
- Percentage: 13.82%
- Funded Amount: \$65.5M

- Amount Received: \$37.3M

This analysis helps identify performing and non-performing loans.

7.4.3 Loan Status Breakdown

The bottom section displays loan performance based on:

- Fully Paid
- Current
- Charged Off

The dashboard compares:

- Loan Applications
- Funded Amount
- Amount Received
- Interest Rate
- DTI Ratio

Loan Status Summary

Loan Status	Applications	Funded Amount	Amount Received
Fully Paid	32.1K	\$351.4M	\$411.6M
Charged Off	5.3K	\$65.5M	\$37.3M
Current	1.1K	\$18.9M	\$24.2M

This section gives a clear understanding of loan repayment performance.



7.5 Overview Dashboard

The **OVERVIEW DASHBOARD** provides detailed multi-dimensional analysis using charts and visualizations.

The dashboard contains slicers for:

- Grade
- Purpose
- Loan Status

Users can filter the entire dashboard interactively.

7.5.1 Monthly Loan Trend

A line and area chart shows monthly loan application trends.

Observation:

- Loan applications increased steadily from January to December.

- December recorded the highest applications.

7.5.2 State-wise Analysis

A filled map chart displays loan applications across different US states.

Observation:

- California had the highest number of loan applications.
- Large states showed higher lending activity.

7.5.3 Loan Term Analysis

A donut chart compares:

- 36-month loans
- 60-month loans

Observation:

- Most borrowers preferred 36-month loan terms.

7.5.4 Employee Length Analysis

A horizontal bar chart analyzes loan applications by employment duration.

Observation:

- Customers with 10+ years of employment formed the largest borrower segment.

7.5.5 Loan Purpose Analysis

Loan applications were analyzed based on purpose such as:

- Debt Consolidation
- Credit Card
- Home Improvement
- Small Business
- Education

Observation:

- Debt Consolidation was the most common loan purpose.

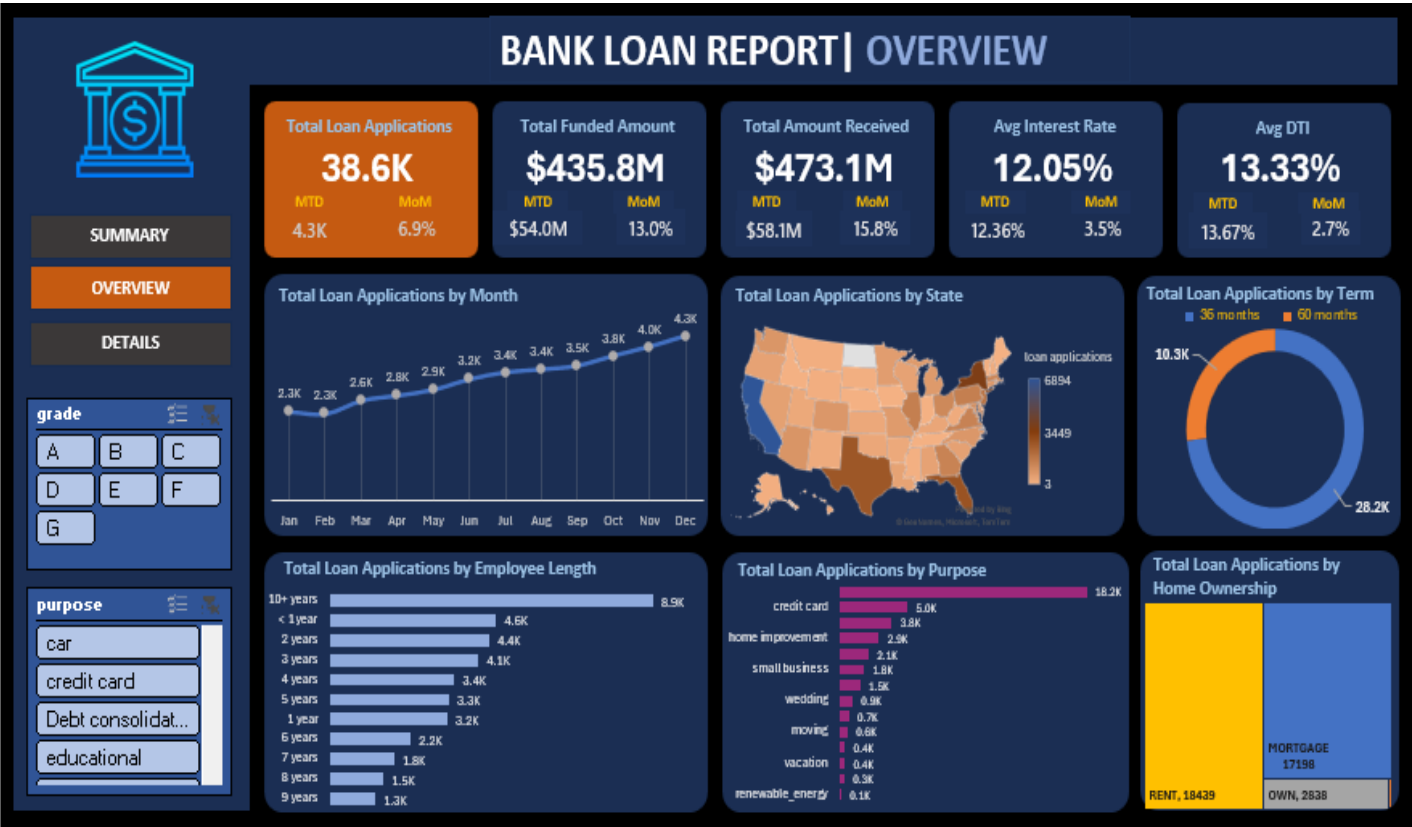
7.5.6 Home Ownership Analysis

A stacked bar chart compares loan applications by:

- RENT
- MORTGAGE
- OWN

Observation:

- RENT and MORTGAGE customers accounted for most loan applications.



7.6 Details Page — Complete Loan Records

The Excel dashboard also includes a separate DETAILS page that displays the complete loan dataset in tabular format.

This page allows users to view detailed information for every loan record after applying filters and slicers from the dashboard. When a user clicks on the DETAILS button from the left navigation panel, the dashboard opens a detailed data table containing all customer and loan-related information.

The DETAILS page includes fields such as:

- Loan ID
- State
- Loan Amount
- Interest Rate
- Loan Status
- Employment Length
- Annual Income
- Loan Purpose
- Home Ownership
- Total Payment

Benefits of the Details Page

- Provides complete record-level information
- Helps users perform detailed loan analysis
- Makes it easy to search and filter customer data
- Supports better decision-making and reporting
- Improves dashboard interactivity and usability

This feature makes the Excel dashboard more user-friendly because users can move from high-level KPI analysis to detailed loan-level information with a single click.

id	address	state	application_type	emp_length	emp_title	grade	home_ownership	issue_date	last_credit_pull_date	last_payment_date	loan_status	Good vs Bad Loan
1E+06	GA		INDIVIDUAL	< 1 year	Ryder	C	RENT	11-02-2021	13-09-2021	13-04-2021	Charged Off	Bad Loan
1E+06	CA		INDIVIDUAL	9 years	MKC Accounting	E	RENT	01-01-2021	14-12-2021	15-01-2021	Fully Paid	Good Loan
1E+06	CA		INDIVIDUAL	4 years	Chemat Technology Inc	C	RENT	05-01-2021	12-12-2021	09-01-2021	Charged Off	Bad Loan
1E+06	TX		INDIVIDUAL	< 1 year	barnes distribution	B	MORTGAGE	25-02-2021	12-12-2021	12-03-2021	Fully Paid	Good Loan
1E+06	IL		INDIVIDUAL	10+ years	J&J Steel Inc	A	MORTGAGE	01-01-2021	14-12-2021	15-01-2021	Fully Paid	Good Loan
1E+06	CA		INDIVIDUAL	3 years	Studio 94 Corp	C	RENT	17-07-2021	16-03-2021	12-08-2021	Fully Paid	Good Loan
1E+06	TX		INDIVIDUAL	10+ years	American Airlines	C	MORTGAGE	19-11-2021	14-06-2021	13-12-2021	Fully Paid	Good Loan
1E+06	PA		INDIVIDUAL	10+ years	SCI Mahanoy	A	OWN	11-06-2021	14-07-2021	14-07-2021	Fully Paid	Good Loan
1E+06	FL		INDIVIDUAL	10+ years	Tech Data Corp	A	MORTGAGE	02-09-2021	15-06-2021	12-10-2021	Charged Off	Bad Loan
1E+06	MI		INDIVIDUAL	10+ years	teltow contracting	B	MORTGAGE	09-02-2021	16-03-2021	16-03-2021	Fully Paid	Good Loan
1E+06	TX		INDIVIDUAL	10+ years	Ericsson	B	MORTGAGE	22-07-2021	13-09-2021	13-08-2021	Fully Paid	Good Loan
1E+06	CA		INDIVIDUAL	3 years	myrvpartsplace.com	B	RENT	11-09-2021	13-03-2021	12-10-2021	Charged Off	Bad Loan
1E+06	CA		INDIVIDUAL	4 years	AEG LIVE	B	RENT	11-08-2021	13-10-2021	13-09-2021	Fully Paid	Good Loan
1E+06	CA		INDIVIDUAL	5 years	henkel corporation	B	RENT	11-12-2021	14-12-2021	14-12-2021	Fully Paid	Good Loan
1E+06	IL		INDIVIDUAL	4 years	AXA Assistance	B	RENT	11-10-2021	14-12-2021	14-12-2021	Fully Paid	Good Loan
1E+06	RI		INDIVIDUAL	< 1 year	HSA-UWC	B	RENT	11-12-2021	14-02-2021	13-10-2021	Charged Off	Bad Loan
1E+06	TX		INDIVIDUAL	1 year	Child's Day	B	RENT	02-12-2021	16-04-2021	14-12-2021	Fully Paid	Good Loan
1E+06	NY		INDIVIDUAL	4 years	OEC Freight	C	RENT	09-10-2021	15-09-2021	12-11-2021	Fully Paid	Good Loan
1E+06	FL		INDIVIDUAL	10+ years	Sandestin Beach Hilton	B	MORTGAGE	12-12-2021	14-12-2021	14-12-2021	Fully Paid	Good Loan
1E+06	MD		INDIVIDUAL	10+ years		D	OWN	02-02-2021	16-05-2021	15-02-2021	Fully Paid	Good Loan
112245	WI		INDIVIDUAL	6 years	Norman G. Olson Insurance	A	RENT	07-07-2021	16-04-2021	10-08-2021	Fully Paid	Good Loan
207910	FL		INDIVIDUAL	< 1 year		A	MORTGAGE	08-01-2021	16-05-2021	10-02-2021	Charged Off	Bad Loan
65426	MI		INDIVIDUAL	< 1 year	Infotrieve, Inc.	B	MORTGAGE	09-08-2021	16-05-2021	11-06-2021	Charged Off	Bad Loan
211723	MD		INDIVIDUAL	4 years	self employed	C	OWN	08-01-2021	09-12-2021	08-08-2021	Charged Off	Bad Loan
662350	NV		INDIVIDUAL	10+ years	Clark County School District	A	MORTGAGE	11-01-2021	13-02-2021	12-08-2021	Charged Off	Bad Loan
1E+06	PA		INDIVIDUAL	6 years	The Mount Sinai School of Medicine	A	MORTGAGE	11-11-2021	13-11-2021	13-06-2021	Charged Off	Bad Loan
405898	GA		INDIVIDUAL	< 1 year	Blue Ridge Design Group	A	MORTGAGE	09-06-2021	16-05-2021	09-12-2021	Charged Off	Bad Loan
445846	MI		INDIVIDUAL	10+ years	Abbott Nutrition	A	MORTGAGE	09-10-2021	16-05-2021	12-02-2021	Charged Off	Bad Loan
659221	MI		INDIVIDUAL	3 years	rice &werthmann	A	MORTGAGE	11-01-2021	16-05-2021	13-10-2021	Charged Off	Bad Loan
782425	UT		INDIVIDUAL	6 years	Five County Assoc of Governments	A	MORTGAGE	11-06-2021	16-05-2021	12-07-2021	Charged Off	Bad Loan
458328	GA		INDIVIDUAL	< 1 year	north ga dentention center	B	MORTGAGE	10-01-2021	10-11-2021	10-07-2021	Charged Off	Bad Loan
744344	WA		INDIVIDUAL	6 years		B	MORTGAGE	11-05-2021	16-05-2021	13-01-2021	Charged Off	Bad Loan

7.7 Dashboard Features

The Excel dashboard includes several interactive features:

- Dynamic slicers and filters
- Automated KPI updates
- Interactive Pivot Charts
- Professional dashboard design
- Easy navigation between sheets

This navigation structure provides a smooth and interactive Business Intelligence experience inside Microsoft Excel.

The dashboard is designed to be user-friendly for both technical and non-technical users.

7.8 Conclusion

The Microsoft Excel Dashboard successfully transformed raw loan data into meaningful visual reports and business insights.

The dashboard provides:

- KPI monitoring
- Loan performance analysis
- Customer behavior insights
- Good vs Bad Loan analysis
- Interactive filtering and reporting

The Excel dashboard serves as an effective Business Intelligence solution and forms the foundation for the advanced Power BI dashboard developed in the next chapter.

CHAPTER 8: POWER BI DASHBOARD DESIGN AND IMPLEMENTATION

8.1 Introduction to Power BI Dashboard

Power BI is a Business Intelligence (BI) tool developed by Microsoft that helps convert raw data into interactive dashboards and visual reports. In this project, Power BI Desktop was used to analyze the bank loan dataset and create a professional dashboard system for loan performance analysis.

The Power BI dashboard was created using the cleaned financial loan dataset containing 38,576 loan records and 25 columns. The dashboard provides interactive filtering, KPI tracking, trend analysis, and detailed loan-level reporting.

The complete dashboard consists of three pages:

1. **Summary Dashboard**
2. **Overview Dashboard**
3. **Details Dashboard**

All dashboards are connected with slicers and filters, allowing users to interact with the data dynamically.

8.2 Data Import and Preparation in Power BI

The cleaned Excel dataset was imported into Power BI Desktop using the **Get Data → Excel Workbook** option.

After importing the dataset, the following preparation steps were performed:

- Verified column data types
- Converted date fields into proper DATE format
- Checked numeric columns for correct decimal formatting
- Created relationships and measures using DAX
- Added calculated columns for analysis
- Built KPI measures for dashboard cards

The dataset was loaded into Power BI successfully and used for all visualizations.

8.3 DAX Measures Used in Power BI

Several DAX (Data Analysis Expressions) measures were created to calculate KPIs dynamically

8.3.1 CORE MEASURES

Total Applications = COUNT (financial_loan[id])

Total Funded Amount = SUM (financial_loan[loan_amount])

Total Amount Received = SUM (financial_loan[total_payment])

Avg Interest Rate = AVERAGE (financial_loan[int_rate]) * 100

Avg DTI = AVERAGE (financial_loan[dti]) * 100

8.3.2 MTD MEASURES

MTD Applications =CALCULATE ([Total Applications], DATESMTD (financial_loan[issue_date]))

MTD Funded Amount = CALCULATE ([Total Funded Amount], DATESMTD (financial_loan[issue_date]))

8.3.3 PMTD MEASURES

PMTD Applications = CALCULATE ([Total Applications], DATEADD(DATESMTD (financial_loan[issue_date]),-1,MONTH))

8.3.4 MoM CHANGE

MoM Applications % = ([MTD Applications] - [PMTD Applications]) / [PMTD Applications]

8.3.5 GOOD / BAD Loan Percentage

Good Loan % =

DIVIDE(

CALCULATE(COUNT(financial_loan[id]),

financial_loan[loan_status] IN {"Fully Paid","Current"}),

COUNT(financial_loan[id])

) * 100

Bad Loan % =

DIVIDE(

CALCULATE(COUNT(financial_loan[id]),

financial_loan[loan_status] = "Charged Off"),

COUNT(financial_loan[id])

) * 100

These DAX measures automatically update whenever slicers or filters are applied.

8.4 Power BI Dashboard Structure

The Power BI report contains three separate pages designed for different levels of analysis.

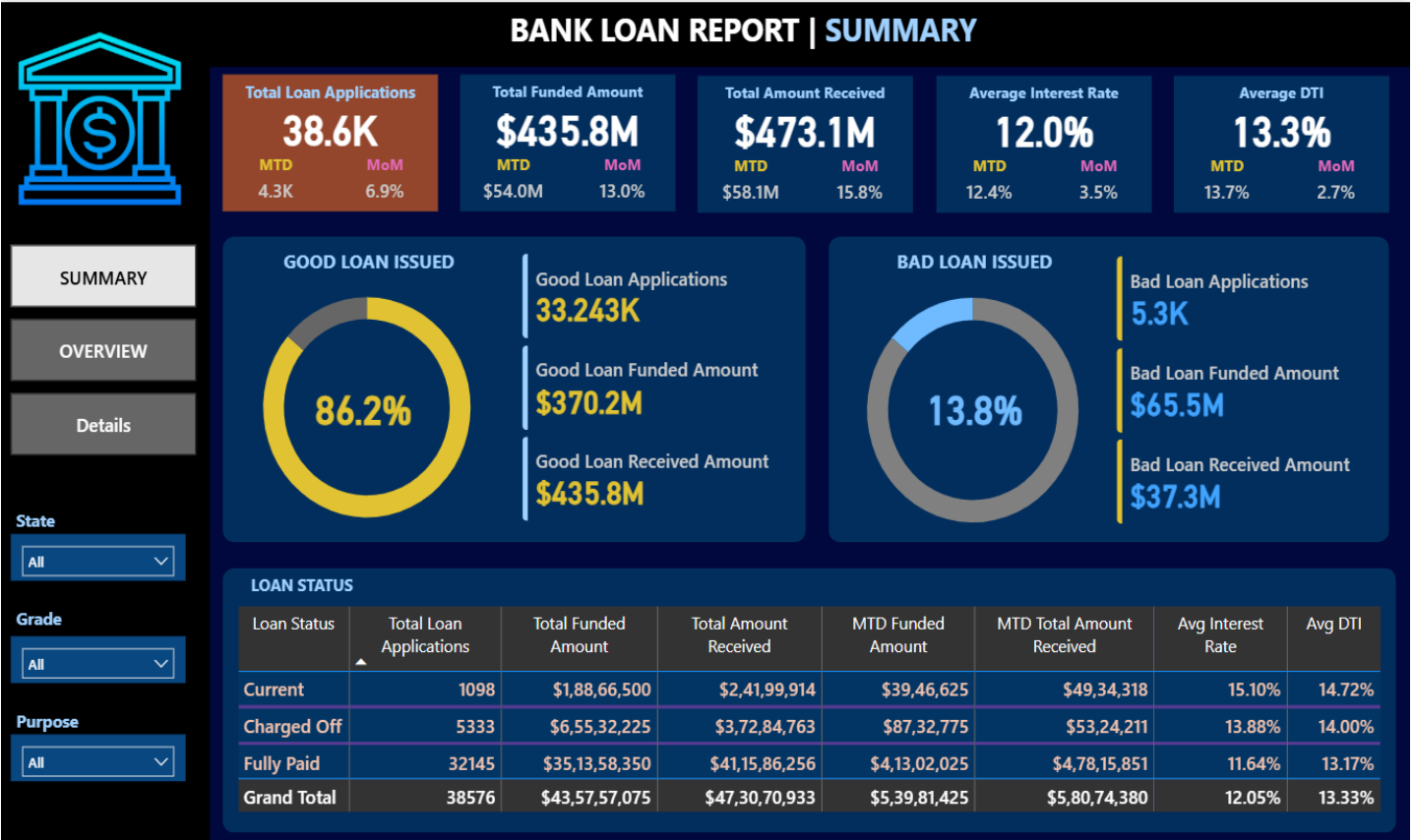
Dashboard Page	Purpose
Summary Dashboard	High-level KPI and loan status analysis
Overview Dashboard	Trend and category-based analysis
Details Dashboard	Complete loan-level detailed records

8.5 Summary Dashboard

The Summary Dashboard provides a high-level overview of the complete loan portfolio using KPI cards, donut charts, and a loan status table. The top section displays five important KPIs: Total Loan Applications, Total Funded Amount, Total Amount Received, Average Interest Rate, and Average DTI. Each KPI card also shows Month-to-Date (MTD) values and Month-over-Month (MoM) growth percentages to help track monthly performance.

The middle section contains Good Loan and Bad Loan donut charts. Good Loans include Fully Paid and Current loans, while Bad Loans represent Charged Off loans. These charts display the number of applications, funded amount, amount received, and percentage share of each category, helping identify portfolio quality and loan recovery performance.

The bottom section includes a loan status table showing detailed metrics for Current, Fully Paid, and Charged Off loans. The dashboard also contains slicers for State, Grade, and Purpose, allowing users to filter all visuals dynamically and analyze specific loan segments interactively.

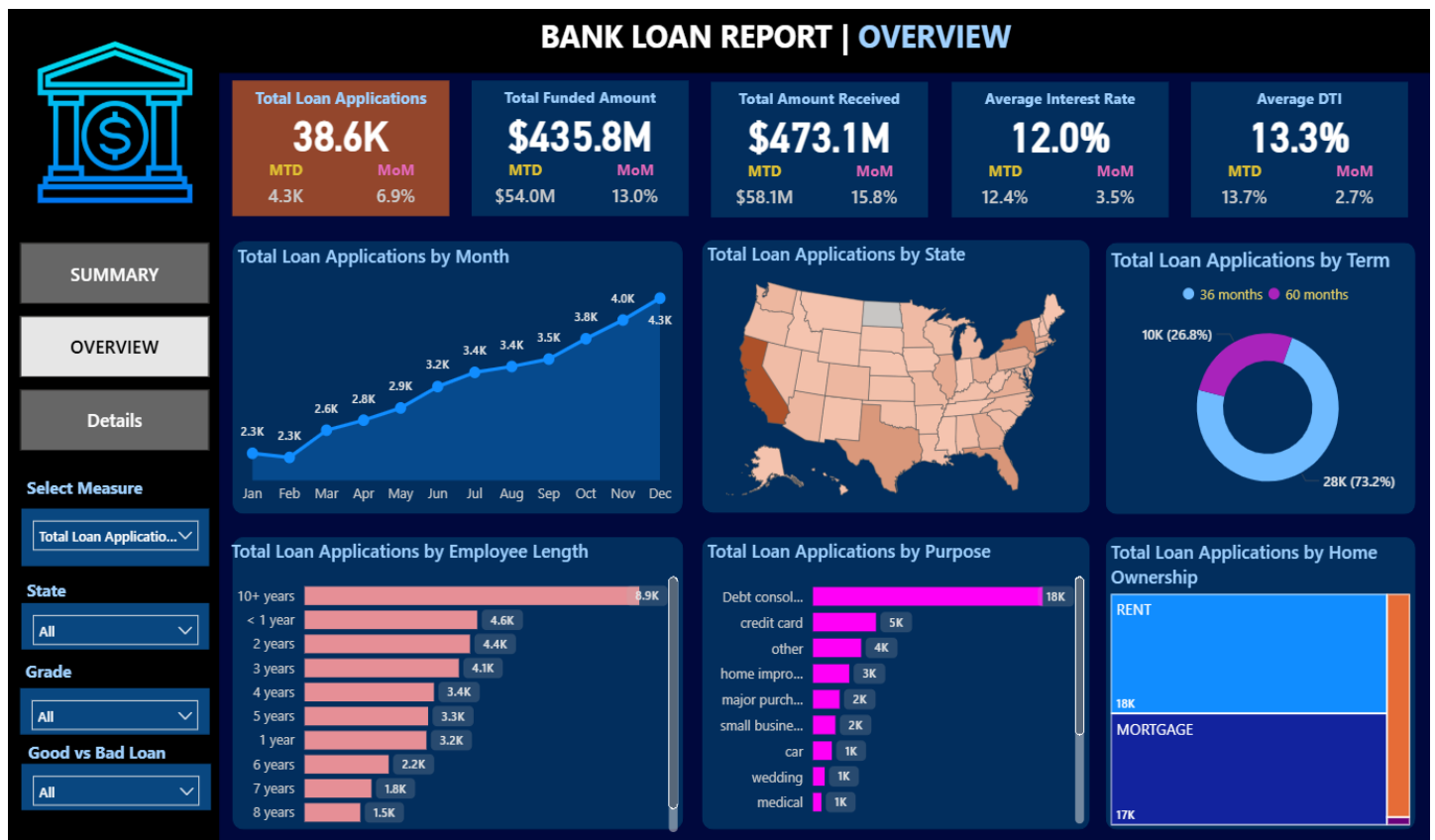


8.6 Overview Dashboard

The Overview Dashboard provides detailed trend and category-based analysis using multiple interactive charts and visualizations. A monthly trend line chart displays the growth in loan applications from January to December, showing a steady increase in lending activity throughout the year.

A filled map visualization is used for state-wise analysis, helping identify geographic regions with higher loan applications. A donut chart compares 36-month and 60-month loan terms, showing that most borrowers prefer shorter repayment periods.

Additional charts analyze loan applications by employment length, loan purpose, and home ownership category. These visualizations help understand borrower behavior, customer profiles, and loan distribution patterns. Interactive slicers for State, Grade, Purpose, and Good vs Bad Loan allow users to filter all charts simultaneously for better analysis and decision-making.



- Monthly Trend Line Chart: Applications grew from 2.3K (Jan) to 4.3K (Dec) — 87% annual increase.
- State Filled Map (Choropleth): California, New York, Texas, and Florida are the highest-volume states.
- Term Donut: 36-month = 73.2% (28K); 60-month = 26.8% (10K).
- Employee Length Bar: 10+ years leads with 8.9K applications; bimodal distribution.
- Loan Purpose Bar: Debt consolidation dominates at 18K (46.7%); credit card second at 5K.
- Home Ownership Treemap: RENT (18K) and MORTGAGE (17K) account for 91%+ of all applications.

8.7 Details Dashboard

The Details Dashboard provides complete loan-level information in a tabular format. It contains detailed fields such as Loan ID, Purpose, Grade, Home Ownership, Issue Date, Funded Amount, Interest Rate, Installment, and Amount Received. This page acts as a detailed reporting section where users can view all individual loan records clearly.

The dashboard also supports interactive filtering and drill-down functionality. When users select any chart, category, or slicer from the Summary or Overview dashboards, the Details page automatically updates and displays only the related records. For example, selecting a particular state, grade, or bad

loan category filters the detailed table accordingly. This feature helps users move easily from summary-level insights to detailed record analysis.



8.8 Slicers and Interactivity

The Power BI dashboard includes interactive slicers for:

- State
- Grade
- Purpose
- Good vs Bad Loan

These slicers are synchronized across all dashboard pages.

Benefits of slicers:

- Real-time filtering
- Better user interaction

- Faster analysis
- Dynamic KPI updates

8.9 Benefits of Power BI Dashboard

The Power BI dashboard provides several advantages:

- Easy understanding of loan performance
- Interactive filtering and drill-down analysis
- Real-time KPI monitoring
- Better decision-making support
- Professional reporting system
- Quick identification of risky loans
- Geographic and trend analysis

The dashboard converts raw banking data into meaningful business insights.

8.10 Conclusion

The Power BI dashboard successfully transforms the raw bank loan dataset into an interactive Business Intelligence reporting system.

Using DAX measures, slicers, KPI cards, charts, maps, and detailed tables, the dashboard provides complete loan analytics in a simple and user-friendly format.

The Summary, Overview, and Details pages together help users analyze:

- Loan growth
- Customer behavior
- Risk distribution
- Geographic performance
- Loan repayment trends

This dashboard demonstrates the practical use of Power BI in banking and financial analytics and serves as an effective decision-support system for loan analysis.

CHAPTER 9: RESULTS AND ANALYSIS

9.1 Introduction

This chapter presents the final results obtained from the Bank Loan Data Analytics project using SQL, Python, Excel, and Power BI. The analysis helps understand loan performance, customer behavior, repayment trends, and risk distribution. The results are displayed through KPIs, charts, dashboards, and detailed analytical reports.

9.2 Overall Loan Portfolio Analysis

The dataset contains a total of **38,576 loan applications**. The overall funded amount is **\$435.8 Million**, while the total amount received from borrowers is **\$473.1 Million**. This shows that the bank recovered more money than the original funded amount due to interest payments.

The average interest rate across all loans is **12.05%**, while the average Debt-to-Income (DTI) ratio is **13.33%**. These metrics help evaluate borrower risk and repayment capacity.

The analysis also shows continuous monthly growth in loan applications and funded amounts, indicating increasing demand for loans during the year.

9.3 Good Loan vs Bad Loan Analysis

Loans were classified into Good Loans and Bad Loans based on loan status.

Good Loans include Fully Paid and Current loans, while Bad Loans include Charged Off loans. The analysis shows that **86.18%** of loans are Good Loans, while only **13.82%** are Bad Loans. This indicates that most borrowers are successfully repaying their loans.

Good Loans account for **33,243 applications** with a funded amount of **\$370.2 Million** and received amount of **\$435.8 Million**. In comparison, Bad Loans account for **5,333 applications** with a funded amount of **\$65.5 Million** and received amount of **\$37.3 Million**.

The lower recovery amount for Bad Loans **highlights the financial loss** caused by loan defaults.

9.4 Temporal and Geographic Insights

Dimension	Key Finding	Business Implication
Monthly Trend	Loan applications grew from 2.3K to 4.3K (Jan–Dec)	Strong lending growth and rising customer demand
December MTD	4,314 applications, \$53.98M funded, \$58.07M received	Positive MoM growth across major KPIs
Regional Analysis	CA recorded highest lending activity followed by major states	Supports regional expansion and market planning
Loan Term	73.2% borrowers preferred 36-month loans	Customers prefer faster repayment and lower interest cost
Employment Length	Highest loans from 10+ years employment group	Employment stability improves loan approval potential
Loan Purpose	Debt Consolidation accounts for nearly 46.7% loans	Customers mainly refinance/manage existing debt
Home Ownership	RENT + MORTGAGE categories contribute 90%+ applications	Loan demand higher among non-property owners

9.5 Key Findings

The major findings of the project are:

- Most loans are successfully repaid, with Good Loans representing over 86% of the portfolio.
- Loan applications increased consistently throughout the year.
- California has the highest loan activity among all states.
- 36-month loans are more preferred than 60-month loans.
- Borrowers with long employment history receive higher funding.

- Debt Consolidation is the most common loan purpose.
- RENT and MORTGAGE borrowers dominate the customer base.
- Interactive dashboards improve analysis and reporting efficiency.

9.6 Conclusion

The project successfully analyzed bank loan data using SQL, Python, Excel, and Power BI. The results provide valuable insights into customer behavior, loan performance, repayment patterns, and financial risk.

The dashboards and analytical reports help transform raw loan data into meaningful business insights, making the system useful for banking analysis and decision-making.

CHAPTER 10: CONCLUSION AND FUTURE WORK

10.1 Conclusion

This project successfully designed and implemented a comprehensive Bank Loan Data Analytics System addressing core operational challenges of loan portfolio monitoring, credit risk classification, KPI reporting, and multi-dimensional data exploration in the banking domain.

The system integrates five technology layers: PostgreSQL (25+ SQL queries), Microsoft Excel (2 dashboard pages with pivot tables), Python/Jupyter Notebook (8 EDA visualizations), and Microsoft Power BI (3-page interactive dashboard with DAX). The pipeline processes 38,576 loan records across 25 attributes, computing all KPIs with Overall, MTD, and MoM time-intelligence, fully cross-validated across all four processing layers.

Key findings: 86.18% good loan rate with 13.82% charge-off rate; \$28.24M credit loss from bad loan under-recovery; 87% annual growth in applications; debt consolidation as dominant purpose (46.7%); geographic concentration in CA, NY, TX, FL.

The project demonstrates how modern Business Intelligence and Data Analytics techniques can support financial institutions in making data-driven decisions, improving risk management, and enhancing operational efficiency. The developed framework is scalable, practical, and highly applicable to real-world banking and financial analytics environments.

10.2 Limitations

Although the project achieved its objectives successfully, certain limitations remain:

- The system mainly performs **descriptive and diagnostic analytics**; predictive analytics and machine learning models for default prediction were not implemented.
- The dataset contains records for only a **single time period**, limiting long-term trend and economic cycle analysis.
- The architecture currently supports **batch data processing** rather than real-time streaming analytics.
- Geographic analysis is restricted to U.S. states and does not support international or multi-currency banking operations.

- External economic indicators such as inflation, unemployment rate, and interest rate fluctuations were not integrated into the analysis.
- The dashboards rely on static dataset refreshes and require manual updates for new incoming data.

10.3 Future Work

- 1. Predictive Modeling:** Integrate logistic regression, random forest, and XGBoost to predict loan default probability at application stage.
- 2. Real-Time Streaming:** Implement Apache Kafka or Azure Event Hubs for live loan data with real-time dashboard updates.
- 3. Advanced DAX:** Add Year-over-Year comparisons, rolling 12-month averages, and cohort analysis.
- 4. Cloud Deployment:** Migrate PostgreSQL to AWS RDS; publish Power BI to Power BI Service with scheduled refresh.
- 5. Excel Power Query:** Add Power Query transformations for automated data refresh and advanced pivoting.
- 6. Automated Regulatory Reporting:** Build HMDA and CRA compliance report automation using the existing data model

REFERENCES

1. [geeksforgeeks-data-analysis](#) — Concepts, applications, and importance of Data Analytics.
2. [skillup-data analytics and business intelligence](#) — Learning resources and theoretical concepts related to Data Analytics and Business Intelligence.
3. [prepinsta- understanding-workflow](#) — Theoretical explanation of analytics workflow and techniques.
4. [GeeksforGeeks Data Analytics Tutorials](#) — Tutorials and theoretical explanations of analytics concepts.
5. [Microsoft Excel](#) — Documentation and Dashboarding Features
6. [W3Schools SQL Tutorial](#) — SQL concepts and database fundamentals used in analytics.
7. [Kaggle Learn](#) — Data Analytics, Python, SQL, and Machine Learning learning resources.
8. [Power BI Documentation](#) — Business Intelligence and dashboarding concepts.
9. [PostgreSQL Documentation](#) — SQL queries, relational database concepts, and data handling techniques.
10. [Pandas Documentation](#) — Data manipulation and analysis concepts in Python.
11. [NumPy Documentation](#) — Numerical computing concepts used in analytics.
12. [Matplotlib Documentation](#) — Data visualization and charting theory in Python.